Air Pollution and Cardiovascular Mortality: Evidence from Thermal Inversions in the U.S.

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Abstract

Thermal inversions are increasingly used as an instrument to estimate the causal effect of air pollution on health, but studies build the inversion measure in different ways and rarely check how these choices affect the estimates. I build and compare thermal inversion instruments across construction choices, using U.S. county and month level data from 2010 to 2023 to study the effect of fine particulate matter (PM2.5) on cardiovascular mortality. I show that the common way of measuring thermal inversions misses about 30 percent of U.S. counties, mostly at higher elevations, while measuring temperature relative to the local ground covers every county. Using this measure, I find that a one-unit increase in PM2.5 raises cardiovascular mortality by 3.8 percent. Results are robust to a range of alternative specifications and pass placebo tests, and I find no evidence of mortality harvesting. Effects are larger among Black individuals than among White and Hispanic individuals, and larger among women, the elderly, low-income counties, and rural counties. These results suggest that PM2.5 has large cardiovascular mortality costs in the United States and that how the instrument is built matters for how these costs are estimated.

Key Words:

Air pollution, Thermal Inversion, Cardiovascular Disease, Mortality, USA

JEL Codes: Q53, I18, C26

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I. Introduction

People breathe low-quality air during outdoor and indoor daily activities, exposing them to different diseases responsible for millions of deaths each year (World Health Organization, 2024). Among air pollutants, fine particulate matter (PM2.5), due to its small size, easily penetrates the human body and has detrimental health impacts (Newby et al., 2015; Pope et al., 2004; Rajagopalan et al., 2018; Sangkham et al., 2024). A large medical literature links PM2.5 exposure to cardiovascular disease (CVD) and finds that both short- and long-term exposure increase cardiovascular morbidity and mortality (Pope et al., 2004). However, much of this evidence does not necessarily identify the causal effect of pollution exposure on cardiovascular mortality. Studying this causal relationship is important as CVD is one of the leading causes of death in the U.S. with substantial economic costs through healthcare expenditures and productivity losses. Thermal inversions are increasingly used to estimate this causal effect, but studies that rely on the same underlying data still construct the inversion measure differently.

In studying air pollution effects, endogeneity issues lead to biased results, making researchers unable to use the Ordinary Least Squares (OLS) method simply. Socioeconomic and behavioral responses, as well as time-varying factors such as economic growth and infrastructure construction, may also affect both pollution and health, leading to omitted variable bias (Currie et al., 2014; Gazze et al., 2024; X. Zhang et al., 2023), and measurement error or reverse causality can bias estimates further. One important step in the air pollution literature is addressing endogeneity through the application of instrumental variables (IV). Thermal inversion, a weather phenomenon that traps pollutants near the ground by limiting vertical dispersion, is among the widely used instruments in the literature (Li & Zhang, 2024; Liu & Ao, 2021; S. Wang et al., 2023). While most prior studies rely on a similar underlying data package

to construct thermal inversions, the process involves several decisions that can produce meaningfully different constructions, and no study has examined how these decisions affect the resulting estimates. This paper addresses that gap by constructing thermal inversions using different data products and methodological choices and evaluating whether these decisions affect the strength of the first stage and the reliability of the causal estimates.

For the purpose of this study, I examine the causal impacts of air pollution exposure on CVD mortality at the monthly county level for the U.S. Using a Two-Stage Least Squares (2SLS) methodology, I instrument air pollution with thermal inversion to address potential air pollution endogeneities. I construct the thermal inversion variable using different data products and methods to compare whether construction methods affect the estimates. To build the instrument, I gather the temperature of the air layers from the National Aeronautics and Space Administration (NASA), and environmental controls and air pollutant levels are collected from the Copernicus Climate Data Store (CDS). I use restricted monthly mortality data from the National Center for Health Statistics (NCHS) for CVD mortality as the outcome of the study. Socioeconomic variables are gathered from the U.S. Census Bureau (CB) and the Bureau of Labor Statistics (BLS). Not all instrumental variable constructions are equally valid, and among the physically valid ones, the preferred construction is the most defensible on conceptual grounds, confirmed by a strong first stage and by greater robustness to violations of the exclusion restriction, following the plausibly exogenous approach of Conley et al. (2012).

Fixed pressure level constructions, the approach used in most prior studies, are measured from sea level and fall below the surface in about 30 percent of U.S. counties. Measuring temperature at 110 and 320 meters above the local surface instead gives a strong instrument in every county. Using this preferred construction, a one-unit increase in PM2.5 raises the

cardiovascular mortality rate by 3.8 percent in the average county. When counties are weighted by population, the estimate is 1.3 percent, which reflects the effect for the average resident. With the fixed pressure level construction, the first stage is about four times weaker, the standard error is about 60 percent larger, and the estimate is about one quarter smaller, although the two estimates are not statistically different from each other. Black individuals experience the largest increase in CVD mortality from PM2.5 exposure among racial groups.

This paper makes two main contributions. First, unlike most prior studies that adopt a single instrument construction, this paper systematically builds and compares thermal inversion instruments across construction choices. It shows that the fixed pressure levels used in most prior studies carry no information about the effect in high-elevation counties and give a much weaker first stage overall, while measuring temperature above the local surface gives a strong instrument in every county. Second, using the preferred construction, it provides new causal estimates of the effect of PM2.5 exposure on cardiovascular mortality for all U.S. counties, using restricted monthly mortality data from 2010 to 2023.

The magnitude of the estimates ranges from 3.8 to 4.6 percent depending on the set of controls and fixed effects, and the plausibly exogenous analysis following Conley et al. (2012) shows the preferred estimate remains significant even if thermal inversions had a hypothetical direct effect on cardiovascular mortality of up to 65 percent of the total reduced form effect. Effects are also larger among women, older adults, lower-income counties, and rural counties. The rural estimate stays the same with population weights, while the urban estimate becomes smaller, which explains why the weighted estimate is smaller than the unweighted one. The results hold up across a range of alternative specifications and placebo tests, with no indication that mortality harvesting explains the estimates. In economic terms, the estimate weighted by

population translates to an annual welfare cost of about $123.6 billion in 2023 based on the EPA's value of a statistical life, providing a policy-relevant benchmark for evaluating air quality regulation.

II. Literature Review

Economists have increasingly focused on the link between air pollution and health outcomes in different age groups. Some literature has covered the impacts of air pollution on early ages, such as child hospitalization rates (Neidell, 2004), low birth weight, and infant mortality (Currie et al. 2009). Another group of literature studied the health impacts of air pollution regardless of the sample age. Giaccherini et al. (2021) show that higher PM2.5 levels have the largest impacts on vulnerable groups such as the elderly, low-educated populations, and migrants.

Different causal methods have been applied to study the health effects of air pollution. Difference-in-differences studies have examined the effects of pollution reduction programs on CVD and non-CVD mortality (Chen et al., 2023) and temperature conditions on CVD and respiratory hospital admissions (Danesh Yazdi et al., 2022). While the DiD method provides useful causal evidence, it does not fully account for unobserved variables and cannot address endogeneity concerns well. Quasi-experimental and regression discontinuity designs based on policy thresholds show causal links between pollution and infant mortality, life expectancy, and physical disability (Chay & Greenstone, 2003; Lavy et al., 2014; Zhao et al., 2023). However, these strategies often rely on localized policy discontinuities or binary treatment cutoffs that are not directly applicable to continuous county-level variation in pollution exposure. Therefore, the

Two-Stage Least Squares (2SLS) approach is a more suitable method for isolating the causal impact of air pollution in this context.

In recent years, some papers have applied instrumental variables to overcome the existing endogeneity in air pollution studies (Arceo et al., 2016). Deryugina et al. (2019) instrument air pollution with local wind direction to assess the impacts of pollutants on mortality and medical expenditures in the U.S. Findings show that higher PM2.5 levels lead to additional deaths among elderly populations. Austin et al. (2023) apply the same instrument to examine air pollution's impacts on COVID-19 mortality and morbidity in the U.S. Gu et al. (2020) by using maximum wind speed show air pollution impacts the mental health of middle and low-income groups in China more than others.

While applied instruments are valid, thermal inversion is a strong IV in the context of this paper to create exogenous variation in pollution exposure. Thermal inversion occurs when a layer of warm air sits above a cooler layer, trapping pollutants close to the ground, which may lead to higher pollution levels. Researchers have applied thermal inversion in different contexts such as labor market efficiencies (Wan & Zhang, 2023; Zhou & Zhang, 2023), infant mortality (Arceo et al., 2016), birth weights (Li and Zhang 2024), medical expenditures (S. Wang et al., 2023; X. Zhang et al., 2023), and a variety of other health-related outcomes. Hicks et al. (2016) instrument for pollution with thermal inversions and find that higher CO levels raise cardiorespiratory and procyclical mortality. Liu and Ao (2021) use insurance data to show that pollution driven by thermal inversions increases respiratory-related healthcare spending. The same instrument has linked pollution to rising obesity levels (Deschenes et al., 2020; G. Zhang et al., 2023).

In addition to the health economics papers, some medical studies have also covered the association between air pollution and CVD in the U.S. (Hart et al., 2015; Mazidi & Speakman, 2018). However, they have neglected the importance of addressing endogeneity concerns and proving the causality between variables.

Prior IV studies on air pollution and health outcomes have largely focused on specific subpopulations or non-CVD outcomes, leaving a gap in causal evidence on PM2.5 effects on cardiovascular mortality at the county-month level in the U.S. Moreover, despite the growing use of thermal inversion as an instrument, existing studies mostly adopt a single construction without examining whether different methodological choices affect the strength of the instrument and the estimates. This paper addresses both gaps using the most up-to-date data from 2010 to 2023. Appendix A provides a comparative overview of the literature on air pollution health effects by method, instrument, and outcome.

III. Data

This study uses data on CVD mortality, PM2.5, temperature of the air layers, weather, and socioeconomic variables at the county-month level from 2010 to 2023. The sample covers 48 contiguous states and the District of Columbia, excluding Alaska, Hawaii, and territories due to data limitations. Table 1 presents descriptive statistics for all variables.

III.a. Cardiovascular Disease Mortality

CVD includes a range of conditions involving the heart and vascular system, such as coronary artery disease, myocardial infarction, and stroke (Thiriet, 2018). Studying CVD mortality as the outcome of the study covers the most important heart and coronary deaths among individuals.

Since air pollution data and environmental variables are available at daily and hourly levels, using mortality data at a higher temporal frequency provides a better match with pollution exposure. However, since the exact date of death is unavailable, the month is the finest temporal unit for analysis. I use the restricted-use mortality dataset from NCHS, which contains county-month-level mortality data along with geographic identifiers, exact month of death, underlying cause of death (ICD-10 codes), demographic characteristics[1], and other medical information[2]. Unlike the public-use mortality data available at the annual level, the restricted files include finer geographic identifiers and precise temporal detail. This provides a more accurate match between pollution exposure and control for spatial and temporal variation in environmental conditions. I restrict the data to deaths where CVD is recorded as the underlying cause of death and define the outcome as the natural logarithm of the mortality rate per 100,000 individuals. Among CVD deaths, White, Black, and Hispanic individuals account for 77%, 14%, and 1.4% of deaths, respectively. Males account for 52% and females for 48%. The majority of deaths are among the older age groups, with 2.4% below age 45, 16.6% for 45-64, 44% for 65-84, and 37% aged 85 and above. As shown in Table 1, the monthly CVD mortality rate has a mean of 28.43 per 100,000 individuals, and the standard deviation of 20.60 shows considerable variation across counties and months.

III.b. Air Pollution

PM2.5 is fine particle matter with a size smaller than 2.5 μm. Based on former studies (Austin et al., 2023; Brook et al., 2002; Hammer et al., 2020) increased level of PM2.5 is associated with cardiovascular events and mortality; therefore, I measure air pollution based on

1. Age, gender, race, marital status, Hispanic origin, education
2. Place of injury, injury at work, method of disposition, autopsy, certifier, tobacco use, pregnancy status, activity code

PM2.5 levels. I gathered the PM2.5 levels from the global reanalysis of the Atmospheric Composition Reanalysis 4 (EAC4)[3] from the Copernicus Atmosphere Monitoring Service (CAMS) gridded at a horizontal resolution of 0.75° × 0.75°. EAC4 reanalysis combines satellite observations and ground measurements with a data assimilation method working with physics and chemistry laws. I aggregate satellite-based data to the county-month level by averaging hourly observations for each county coordinate. The average county experiences a monthly PM2.5 concentration of 10.11 $\mu g/m^3$ with a standard deviation of 4.56. The average county-year has a mean PM2.5 concentration of 10.11 $\mu g/m^3$ at the annual level, below the EPA's annual standard of 12 $\mu g/m^3$ in effect over the sample period, while 14.6% of county years exceed this threshold[4].

III.c. Thermal Inversion

While on a typical day air temperature decreases with altitude, during a thermal inversion, the air near the ground is cooler than the air above and cannot rise and mix with clean air, causing pollutants to become trapped close to the surface (Xia et al., 2022). Environmental phenomena can cause thermal inversion in coastal areas and valleys through topography, or on a clear night if the air layer near the surface rapidly loses heat whereas the air layer above retains the heat from the ground (Liu & Ao, 2021).

A thermal inversion occurs when the temperature of an upper atmospheric layer exceeds the temperature of the layer below it, and the magnitude of this temperature difference measures

3. ECMWF Atmospheric Composition Reanalysis 4
4. The summary statistics table shows a maximum of 211 $\mu g/m^3$ for PM2.5 concentration in Baker County, Florida, in June 2011, and a second-highest value of 208.87 $\mu g/m^3$ in Ware County, Georgia, in the same month. Both reflect smoke from the Honey Prairie Complex Fire, a large wildfire in the Okefenokee Swamp that burned for months in 2011 (NASA Earth Observatory, 2011; New Georgia Encyclopedia, 2026)

inversion strength. Constructing thermal inversion requires making several choices: which atmospheric levels to compare and from which data product, whether to measure inversion as occurrence or as a continuous strength measure, and how grid-level readings are mapped to counties. I systematically compare these choices to evaluate whether construction decisions affect instrument strength and the reliability of the causal estimates. Figure 1 summarizes the full construction process and the resulting instruments.

III.c.1. Data and Level Selection

Constructing a thermal inversion requires selecting two atmospheric levels to compare temperatures and look for an inversion condition. I use atmospheric temperature data from MERRA-2, the Modern Era Retrospective Analysis for Research and Applications Version 2, produced by NASA, and compare suggested level selections. I collect two MERRA-2 data products and compare approaches to level selection within each.[5]

III.c.1. a. Pressure level product

The pressure level product[6] is a three-dimensional product that provides instantaneous temperature readings for 42 pressure levels at six-hour intervals on a 0.5° × 0.625° global grid, with grid cells covering approximately 56×58 km. Pressure levels are atmospheric layers identified by their air pressure, where lower pressure values belong to higher altitudes.

Not all pressure levels are suitable for constructing thermal inversion instruments, and level selection matters for two reasons that depend on the geographic scope and terrain characteristics of the study. The first reason is terrain coverage. Most prior studies use the 1000 and 975 hPa pressure levels, corresponding to approximately 110 and 320 meters above sea level

5. An alternative ground based source for upper air temperature is NOAA's Integrated Global Radiosonde Archive (IGRA). Only 94 U.S. stations have continuous data spanning the full 2010-2023 study period, far short of the coverage needed for a national county-month panel.
6. M2I6NPANA version 5.12.4. Available at: https://disc.gsfc.nasa.gov/datasets/M2I6NPANA_5.12.4/summary

(Deschenes et al., 2020; Li & Zhang, 2024; S. Wang et al., 2023; G. Zhang et al., 2023; X. Zhang et al., 2023). However, these levels are measured above sea level and fall below the local surface terrain in high-elevation areas. A representative sample of days shows that approximately 47 percent of grids and 30 percent of counties have invalid temperature data at both the 1000 and 975 hPa levels. This layer pair produces valid instruments in specific settings, such as at the weekly level for Georgia, a mostly flat state where terrain coverage is not a concern (Ghafarianghadim & Heutel, 2026), and in individual-level weekly data for a selected sample of counties with EPA monitoring stations (You & Chou, 2025). However, these levels are not well suited for a monthly sample of all counties due to terrain coverage issues. Using the 975 and 950 hPa levels instead, corresponding to approximately 320 and 540 meters above sea level, reduces the share of affected grids to 36.3 percent (the share of fully affected counties to 18.2 percent), though these counties remain concentrated in the western United States and the Appalachian corridor. Given the remaining terrain coverage problem in a national sample covering 3,107 counties, section V.b compares the preferred construction with the 1000 and 975 hPa construction.

The second reason relates to Planetary Boundary Layer Height (PBLH). The planetary boundary layer is the lowest part of the atmosphere where surface heating drives active mixing of air and pollutants, and PBLH measures the height of this mixing layer (Barbara J. Finlayson-Pitts, 1986). Pollutants emitted at the surface are contained within this layer, and a thermal inversion forming inside this layer suppresses the normal mixing, trapping pollution near the surface. An inversion forming above the PBLH would not affect surface pollution because normal mixing below it continues uninterrupted. The average PBLH for my sample is 568

meters, with a mean minimum of 552 meters, so selected atmospheric levels should lie within this layer to capture inversions that trap surface pollution.

To address the terrain coverage problem while retaining the pressure level data product, I construct an adaptive layer specification (the first left box in the top row of Figure 1) that identifies the lowest valid pressure layer at each grid point and computes the temperature difference relative to the next level above it. This fully eliminates the terrain coverage problem by always using the lowest available level at each location. Adaptive construction may not capture a consistent atmospheric layer across locations, as different grid cells use different pressure levels. This cross-grid inconsistency motivates the use of model-level data as an alternative construction.

III.c.1. b. Model-level data

The MERRA-2 model level product reports temperature every three hours at 72 vertical layers defined relative to the local surface, and every grid has valid temperature data regardless of local elevation[7]. I interpolate temperatures to 110 and 320 meters above local ground at each grid point. These heights match the physical targets of the 1000 and 975 hPa pressure levels used in prior studies, but because model level data measures height above the local surface, every grid has valid temperature data regardless of elevation. I also construct instruments at 326 and 547 meters above local ground to examine whether results are sensitive to the choice of height pair within the boundary layer. Both height pairs sit inside the average planetary boundary layer height of 568 meters, making both suitable for capturing inversions that trap surface pollution. The results section compares estimates across adaptive and model-level constructions and across

7. M2I3NVASM version 5.12.4. Available at: https://disc.gsfc.nasa.gov/datasets/M2I3NVASM_5.12.4/summary

height pairs to assess the robustness of the identification strategy. Using these data products and level selections, I construct inversion occurrence and inversion strength as two distinct inversion measures:

III.c.2. Inversion Measures

I construct two types of inversion measures from each of the level specifications described above: inversion occurrence, capturing whether an inversion happened, and inversion strength, capturing the magnitude of the temperature difference when an inversion occurs.

III.c.2.a. Inversion Occurrence

To build inversion occurrence instruments, two choices of daily aggregation (count and binary) and spatial mapping (zonal and IDW) apply to each of the three level selections. Combining these two choices across the three level selections yields twelve occurrence instruments in Figure 1.

The first choice is defining the daily value at each grid point. The pressure level product used for adaptive level selection has four six-hour intervals per day, while the model level product has eight three-hour intervals per day. Following Deschenes et al. (2020), the count approach records the number of intervals that experienced an inversion during the day. The daily value ranges from zero to four for the pressure level product and from zero to eight for the model level data. Following X. Zhang et al. (2023) and Liu & Ao (2021), the binary approach assigns a value of one to a day if at least one interval during the day has an inversion and zero otherwise. Daily values are then summed across all days in the month to obtain a monthly grid value.[8]

8. Both sum and averaging produce near-identical instruments since the only difference is scaling by the number of days in the month. I report results on instruments with summed value for grid month.

The second choice is the mapping approach from grids to counties. The zonal statistics approach takes the mean of all grid cells overlapping the county polygon. The inverse-distance weighted approach takes a weighted average of all grid points within 100 kilometers of the county centroid (You & Chou, 2025).

Walking through Figure 1, I check the inversion condition at each interval. This step depends on the data product and the selected levels: for adaptive levels, whether the next valid level above is warmer, and for model levels, whether the temperature at the upper level exceeds the temperature at the lower level. I count the number of intervals meeting the inversion condition over a month, for each grid and then apply zonal or IDW mapping to the county-month value. This process applied to adaptive level selection yields constructions 1 to 4; applied to model level selection at 110 and 320 meters yields constructions 5 to 8; and applied to model level selection at 326 and 547 meters yields constructions 9 to 12. The binary approach follows the same structure but flags the entire day as an inversion day if at least one interval meets the inversion condition.

Panels A, B, and C of Table 2 report descriptive statistics for inversion occurrence instruments under adaptive pressure level selection, model level selection at 110 and 320 meters, and model level selection at 326 and 547 meters, respectively. The average county experiences about 33 inversion intervals per month under adaptive level selection and about 31 under model level selection at 326 and 547 meters, while the same county experiences about 77 inversion intervals per month under model level selection at 110 and 320 meters. The higher count at 110 and 320 meters reflects that this height pair sits closer to the surface and detects more frequent inversions, while the 326 and 547 meters pair captures only inversions strong enough to reach that height. Binary instruments record the number of days in a month where at least one

inversion interval occurred. The average county has about 23 inversion days per month under 110 and 320 meters selection and about 11 inversion days per month under 326 and 547 meters selection, consistent with the count-based pattern. Zonal and IDW mapping produce near-identical means within each panel, confirming that the mapping choice does not substantially affect the instrument.

III.c.2.b Inversion Strength

Figure 1 also shows the construction of inversion strength instruments using the magnitude of the temperature difference. For each interval, I record the positive temperature difference between the upper and lower levels when an inversion is present and assign zero to non-inversion intervals. I then average across all intervals in the day to obtain a daily grid value (across four intervals for pressure level data and eight intervals for model level data). Then, I average daily values across all days in the month to obtain a monthly grid value. Applying zonal statistics and IDW mapping gives two strength instruments per level selection and six in total.

Panels D, E, and F of Table 2 report descriptive statistics for inversion strength instruments under adaptive pressure level selection, model level selection at 110 and 320 meters, and model level selection at 326 and 547 meters, respectively. The strength measure records the average temperature difference in degrees Celsius between the upper and lower levels across all intervals, with intervals without an inversion counted as zero. Adaptive selection in Panel D shows a mean of 0.54°C, and model-level instruments at 326 and 547 meters in Panel F show an average of 0.16°C. Model-level instruments at 110 and 320 meters in Panel E show a higher mean of 0.52°C. The higher mean at this height pair reflects that inversions detected closer to the surface involve larger temperature gradients between adjacent layers than inversions measured at higher levels. Zonal and IDW mapping produce near-identical means within each panel,

confirming the mapping choice does not drive instrument variation. These decisions provide 18 different constructions of thermal inversion and section V.a discusses the strategy for selecting among them.

III.d. Environmental Control

Data on temperature, evaporation, and wind speed are obtained from the CAMS EAC4, and precipitation is from ERA5[9] reanalysis, available through the Copernicus Climate Data Store (CDS). CAMS provides global atmospheric composition and meteorological data at a 0.75° × 0.75° grid, and ERA5 provides data at a 0.25° × 0.25° grid (CDS, 2024). I aggregate the daily data to the monthly level for counties based on the U.S. Census Bureau County shapefiles.

III.e. Socioeconomic Controls

The share of the population with at least a high school diploma as the education variable, and median income are gathered from the U.S. Census Bureau. On average, 76 percent of the population holds at least a high school diploma, and the median income is 51,097 dollars. The monthly unemployment rate is collected from the BLS with a county monthly average of 5.70.

III.f. Placebo Outcomes

To validate the instrument, I construct monthly county-level counts of homicide and accidental falls mortality from NCHS mortality records and convert them to rates per 100,000 residents. These outcomes have no plausible biological pathway from short-term PM2.5 exposure and

9. ERA5 is the fifth generation European Centre for Medium-Range Weather Forecasts (ECMWF) reanalysis for the global climate and weather for the past 8 decades. Data is available from 1940 onwards. ERA5 replaces the ERA-Interim reanalysis.

serve as placebo outcomes in the reduced form analysis[10]. The average county-month homicide rate is 0.363 per 100,000 residents, and the average accidental fall rate is 1.396 per 100,000 residents.

IV. Empirical Strategy

IV.a. Two-Stage Least Squares Method (2SLS)

The existing endogeneity in air pollution studies makes studying the causal effect of air pollution on CVD mortality challenging. Measurement error, omitted variables bias, and the possibility of reverse causality make OLS a biased estimator with unreliable results.

I use thermal inversion as the IV for air pollution to address endogeneity concerns. Thermal inversion traps air pollutants near the ground and limits their dispersion, showing a correlation between thermal inversion and air pollution, which satisfies the relevance condition. Furthermore, thermal inversion is uncorrelated with unobserved factors affecting CVD mortality, and it affects the outcome variable only through air pollution which plausibly satisfies the exclusion restriction. By using thermal inversion as the IV, I adopt the following 2SLS empirical strategy to follow the goal of the study:

$$P_{jt} = \alpha_0 + \alpha_1 Z_{jt} + \alpha_2 W_{jt} + \delta_j + T_t + \mu_{sm} + \tau X_{jt} + \varepsilon_{jt} \quad (1)$$

$$ln(Y_{jt}) = \beta_0 + \beta_1 \hat{P}_{jt} + \beta_2 W_{jt} + \delta_j + T_t + \mu_{sm} + \theta X_{jt} + u_{jt} \quad (2)$$

Equation (1) estimates the effect of thermal inversion on air pollution in the first stage, and Equation (2) uses instrumented air pollution in the second stage to estimate the causal effect on the outcome. Due to the availability of mortality data at the monthly level, I estimate both

10. To address extreme values driven by very small county populations, both rates are winsorized at the 99th percentile.

equations at the county-month level to ensure all variables are measured consistently. P_{jt} is PM2.5 in the county j at year-month t. Y_{jt} is the outcome of interest, measuring the CVD mortality rate in the county j at year-month t. Z_{jt} is the county-month level thermal inversion instrument constructed from atmospheric temperature data as described in Section III. W_{jt} includes weather controls: wind speed, temperature, evaporation, and precipitation. δ_j represents county fixed effects capturing time-invariant differences across counties in geography, demographics, and baseline health conditions. T_t represents year fixed effects absorbing national annual shocks. μ_{sm} represents state×month fixed effects absorbing regional seasonal variation in both thermal inversion and cardiovascular mortality. X_{jt} is a vector of socioeconomic controls including the share of population with at least a high school diploma, median household income, and the unemployment rate. Standard errors are clustered at the county level to account for serial correlation and within-county dependence. The coefficient β_1 measures the causal effect of PM2.5 on the CVD mortality rate. All main regressions give each county equal weight, so β_1 measures the effect in the average county. I use this as the main specification because the paper compares how well each instrument construction covers U.S. counties. When counties are weighted by population, β_1 measures the effect for the average resident instead. I report these weighted estimates in Tables 6 and 10 and use them for the national welfare cost in Section V.f as that cost depends on the number of people affected.

IV.b. Instrument Selection Procedure

I select a preferred instrument using a two-step procedure among the 18 candidate constructions. First, within each family, I evaluate aggregation and mapping using conceptual reasoning alone, without reference to mortality outcomes, then confirm the choice with first-stage

performance. This avoids specification searching, since choosing the construction that produces the largest or most significant coefficient would make the reported estimate and its inference unreliable. Between count and binary aggregation, count keeps the frequency of inversion events, while binary only records whether an inversion happened at all, treating a county-month with one inversion the same as a county-month with many. I prefer count over binary for this reason. Between zonal and IDW mapping, zonal keeps the instrument tied to conditions inside the county, while inverse-distance weighting also pulls in grid points up to 100 kilometers outside the county. I prefer zonal for the same reason, even where IDW gives a somewhat higher F-statistic. Among the three level selections, adaptive selection uses pressure level data, which assigns a different atmospheric level to each grid point depending on local elevation. Model level data instead measures a fixed height above the surface at every grid point regardless of elevation, giving a more consistent atmospheric layer across locations. Within model level data, I prefer 110 and 320 meters over 326 and 547 meters because it is closer to the surface, where pollution exposure is concentrated. Constructions at 326 and 547 meters serve as a check on sensitivity to the height pair rather than as a competing choice. With aggregation, mapping, and level selection fixed this way, I then check the first-stage F-statistic, which confirms this construction produces a strong instrument rather than serving as the basis for choosing among alternatives. The inversion strength family involves no count-binary choice, since it measures the magnitude of the temperature difference directly, so only mapping and level selection apply. The same reasoning favors zonal mapping and model level data at 110 and 320 meters for strength as well. Fixed pressure levels are not among the candidate constructions. Because they are measured from sea level, the 1000 and 975 hPa levels used in most prior studies fall below the surface in about 30 percent of counties, so

the instrument has no valid reading there. Section V.b compares the preferred construction with this fixed pressure level approach and shows how much the choice matters for the estimates.

Second, after identifying one preferred occurrence construction and one preferred strength construction, I compare their robustness to potential exclusion restriction violations using the plausibly exogenous framework of Conley et al. (2012). This step uses mortality data through the reduced form, but it does not depend on the size or significance of the 2SLS estimate. The exclusion restriction assumes thermal inversions affect cardiovascular mortality only through PM2.5 and have no direct effect on the outcome. While this assumption is standard in the thermal inversion literature and is supported by the physical mechanism described in Section III, it cannot be formally tested. Conley et al. (2012) argue that most instruments are only plausibly exogenous and propose methods for assessing how sensitive results are to small violations of this assumption. I apply their approach to compare how sensitive the two preferred constructions are to potential violations of the exclusion restriction, rather than as a formal test of whether either instrument fully satisfies it. Equation 2 assumes thermal inversions have no direct effect on mortality and that their impact operates exclusively through PM2.5. Following Conley et al. (2012), I relax this by allowing the instrument to enter the structural equation directly:

$$ln(Y_{jt}) = \beta_0 + \beta_1 \hat{P}_{jt} + \gamma Z_{jt} + \beta_2 W_{jt} + \delta_j + T_t + \mu_{sm} + \theta X_{jt} + u_{jt} \quad (3)$$

where γ captures the direct effect of thermal inversions on mortality through channels other than PM2.5, which equals zero under the standard exclusion restriction. I apply the Union of Confidence Intervals method from Conley et al. (2012) using the plausexog command in Stata (Clarke & Matta, 2018)[11]. I set the lower bound of γ to zero because any direct effect of thermal

11. The plausexog command follows ivregress syntax and does not support the absorb option for high dimensional fixed effects. Including county, year, and state×month fixed effects directly as dummy variables would require estimating over 1,800 indicator variables simultaneously. Therefore, I portalled out county, year, and state×month fixed effects and all control variables from the outcome, endogenous variable, and each instrument prior to estimation

inversions on cardiovascular mortality is likely to operate in a positive direction. As the upper bound, I use each instrument's reduced form coefficient, which captures the total effect of the inversion on mortality through all channels. For each construction, I find the breakpoint, defined as the value of γ at which the lower confidence bound on the PM2.5 coefficient first crosses zero, meaning the estimate loses significance, and express it as a share of the reduced form coefficient. A higher breakpoint percentage means the estimate survives a larger hypothetical direct effect, so the construction is less fragile to violations of the exclusion restriction. I report breakpoints for all 18 constructions in Table 3 and use them to choose between the preferred occurrence and preferred strength constructions.

V. Results

V.a. Selecting the Preferred Instrument

I present findings separately for inversion occurrence and inversion strength instruments, since these measures capture different aspects of atmospheric conditions and are not directly comparable. Occurrence instruments measure whether and how frequently inversions occur, while strength instruments measure the magnitude of the temperature difference during inversion episodes. I identify a preferred construction within each family, then assess which of the two provides the more reliable source of exogenous variation.

Figure 2 provides a graphical summary of F-statistics for all 18 valid constructions and highlights the preferred construction within each instrument family. Aggregation and mapping are evaluated separately from level selection because they represent different dimensions of

using reghdfe. By the Frisch-Waugh-Lovell theorem, the 2SLS coefficient on PM2.5 is numerically identical whether fixed effects are absorbed directly or partial out beforehand.

instrument construction. Count and binary aggregation respond to height pairs in opposite directions, so comparing height pairs before fixing aggregation would not have a clear answer. Count aggregation preserves information on inversion frequency, whereas binary aggregation records only whether an inversion occurred. Consequently, binary aggregation assigns equal weight to county months with a single inversion event and county months with many inversion events, whereas count aggregation allows more frequent inversion activity to contribute proportionally more variation. Zonal mapping restricts the instrument to atmospheric conditions within county boundaries, but IDW incorporates locations up to 100 kilometers away. While IDW produces higher F-statistics in most level selections, I select zonal as it keeps the instrument tied to conditions inside the county, and both constructions are well above the conventional thresholds regardless of which is chosen. Based on these conceptual considerations, I adopt count and zonal mapping as the baseline construction, then compare alternative level selections against it. With aggregation and mapping fixed, I compare level selection and height pair based on the F-statistics of the count-zonal construction. Adaptive selection gives 399.39, model level at 110 and 320 meters gives 462.53, and model level at 326 and 547 meters gives 224.04. Among occurrence constructions, the count instrument at 110 and 320 meters with zonal mapping is the one selected on conceptual grounds. Its first-stage F-statistic of 463 shows that it is a strong instrument, well above conventional thresholds. A strong first stage shows that the instrument moves PM2.5, but it does not by itself show that the instrument is valid, so I do not use the F-statistic to rank constructions. The placebo, plausibly exogenous, and robustness tests in the following sections address validity.

The strength family follows the same procedure, with mapping and level selection as the only choices. For the same reasons as for occurrence, I prefer zonal mapping and model level

data at 110 and 320 meters, which measures a consistent layer above the surface at every grid point. The preferred strength construction has a first-stage F-statistic of 469.

The preferred occurrence and strength constructions both satisfy the relevance condition and exhibit very strong first-stage relationships with PM2.5. Because occurrence and strength measure different aspects of thermal inversions, their first-stage coefficients and F-statistics are not directly comparable and are therefore not used to rank one measure against the other. Instead, after identifying a preferred construction within each family, I evaluate which provides the more reliable instrument in this setting.

The preferred occurrence construction exhibits substantially greater stability across plausible height selections than the preferred strength construction. Changing the height pair from 110 and 320 meters to 326 and 547 meters reduces the first-stage F-statistic from 462.53 to 224.04 for occurrence but from 469.05 to 18.79 for strength, indicating that occurrence measures are considerably less sensitive to the choice of atmospheric levels. This difference is suggestive, but it is not by itself a decision rule, so I turn to a different source of evidence to make the final selection.

Table 3 reports plausibly exogenous breakpoints following Conley et al. (2012) for all valid constructions. The breakpoint is the share of the reduced-form coefficient that the direct effect of the instrument on mortality must reach before the second-stage estimate loses significance; a higher breakpoint indicates greater robustness to potential exclusion restriction violations. This measure uses mortality data through the reduced form but does not depend on the magnitude or statistical significance of the 2SLS estimate itself. Comparing the two preferred constructions, occurrence reaches a breakpoint of 65 percent, while strength reaches only 5 percent; this is consistent with the pattern described above. Based on its greater robustness across

height selections and its higher breakpoint, I select the count-based occurrence instrument at 110 and 320 meters with zonal mapping as the preferred specification.[12]

V.b. Estimates Across Instrument Constructions

Table 4 reports results using count and binary occurrence instruments. Constructions 1 to 4 use adaptive pressure level selection, constructions 5 to 8 use model level data at 110 and 320 meters above local ground, and constructions 9 to 12 use model level data at 326 and 547 meters above local ground. All twelve occurrence instruments produce positive and significant second-stage estimates. Count instruments give estimates between 0.027 and 0.042 across all level selections and mapping choices, while binary instruments give larger estimates, between 0.036 and 0.087. Mapping choice has little effect on the estimate under any level selection. The preferred instrument in column 5 gives an estimate of 0.038, close to its IDW counterpart in column 6 (0.033). As a diagnostic check, an overidentification test using the three count-zonal instruments jointly rejects equality at the 5 percent level but not at the 1 percent level (Hansen J = 6.29, p = 0.043). Given the large sample and strong instruments, the test detects small differences between estimates that are similar in size, ranging from 0.027 to 0.042. This test is used only to compare constructions, not to combine instruments in the main specification.

Table 5 reports results using inversion strength instruments. Most strength instruments have strong first stages but produce small second-stage estimates. Adaptive strength gives

12. These 18 constructions are not independent instruments. Each measures the same inversion event with a different data product, height pair, aggregation rule, or mapping method, so they are positively correlated. Zonal and IDW versions of the same construction correlate at 0.97 or higher, and count and binary versions at the same level selection correlate at 0.75 to 0.91. Correlations are lower across level selections, at 0.45 to 0.81 for count occurrence and 0.54 to 0.86 for strength, since higher layers capture only deeper inversions. Since these are not separate sources of variation, I evaluate each construction choice to pick a preferred one per family based on conceptual validity and first-stage strength, instead of pooling all of the instruments.

estimates of 0.006 and 0.007, and model level strength at 110 and 320 meters gives 0.013 and 0.011, the first significant only at the 10 percent level. Only strength at 326 and 547 meters gives larger and significant estimates (0.058 and 0.056), but these instruments are weak, with first-stage F-statistics near 20. For adaptive strength, the reduced-form relationship with CVD mortality is small and insignificant (Anderson-Rubin p-values of 0.41 and 0.34), so inversion strength shifts PM2.5 but shows little relationship with mortality in this setting. Occurrence and strength instruments measure different aspects of thermal inversion, and their magnitudes are not directly comparable.

These comparisons come after the selection in Section V.a and are not used to choose the instrument. Table 6 tests whether the terrain coverage problem is a meaningful concern for this paper's findings, comparing the preferred instrument with a similarly constructed pressure-level alternative (count-based, zonal mapping at 1000 and 975 hPa). I split the sample into 953 high-elevation counties, where most overlapping grid cells have no valid temperature at both pressure levels, and 2,154 low-elevation counties. The preferred model-level instrument produces stable, significant estimates in both subsamples, 0.041 in high-elevation counties and 0.033 in low-elevation counties, with first-stage F-statistics of 92 and 478. The pressure-level counterpart has no first stage in high-elevation counties, so it carries no information about the effect of PM2.5 in about 30 percent of counties. Although it produces a coefficient close to the preferred estimate in the full sample (0.028 compared with 0.038), this estimate comes almost entirely from low-elevation counties and is less precise, with a first-stage F-statistic of 110 compared with 463. In low-elevation counties it gives a smaller estimate of 0.014, significant only at the 10 percent level, since the 1000 hPa level also falls below the surface in parts of this group. The other fixed pressure level constructions show a similar pattern (Appendix Table B1).

The difference between the two constructions becomes clearer when each county is weighted by its population (Panel B of Table 6). The model level instrument still gives positive and significant estimates in the full sample and in both high and low-elevation counties. The fixed pressure level instrument gives positive but insignificant estimates in all three samples, reflecting that the instrument based on pressure level data is too imprecise to identify the effect for the average resident.[13]

To assess how construction choices affect comparisons with the literature, I compare my estimates with studies that instrument air pollution with thermal inversions using the same NASA MERRA-2 reanalysis product. These studies differ in time period, geographic level, country, and outcome, making direct comparison difficult. S. Wang et al. (2023) count inversions using the lowest fixed MERRA-2 pressure levels, Liu & Ao (2021) use a binary measure at the same levels, and You & Chou (2025) use a count measure mapped with IDW. Replicating these fixed-level constructions in my data gives positive estimates of 0.028 for count, 0.046 for binary, and 0.037 for count with IDW mapping (Table 6 and Appendix Table B1), consistent in sign with these studies and with the model-level versions at the same nominal heights (0.038, 0.087, and 0.033 for constructions 5, 7, and 6). The difference between the two approaches is therefore not the sign of the estimate but its coverage and precision. In a national U.S. sample, the fixed-level constructions have much weaker first stages and identify the effect mostly from low-elevation counties. Arceo et al. (2016) build their measure from daily balloon soundings that record temperature directly at each altitude, and despite this difference in

13. Population weighting reduces precision because a small number of large counties carry most of the weight, so the estimates rely on far fewer independent observations than the 3,107 counties in the sample. This is especially true among high-elevation counties, where most of the population lives in a few metropolitan areas. For this reason, the first-stage F-statistics fall under weighting for both instruments, even though the first-stage coefficients do not become smaller.

construction and setting, their estimate of 0.23 on infant mortality is consistent in sign with my findings.

V.c. Effect of PM2.5 on Cardiovascular Mortality

Table 7 shows the impact of PM2.5 on CVD mortality rate using the count-based occurrence instrument at 110 and 320 meters above ground with zonal mapping. All columns include county fixed effects absorbing time-invariant county differences and state×month fixed effects absorbing regional seasonal patterns. Column 1 includes year fixed effects but no control variables. The positive and significant second-stage estimate confirms the instrument identifies a causal effect even without controls. Columns 2 and 3 add socioeconomic controls and weather controls separately to isolate which set of controls drives the change in identification strength. Adding socioeconomic controls alone (column 2) leaves the F-statistic essentially unchanged, from 256 to 261, since county-level measures such as income, unemployment, and education vary slowly over time and are only weakly correlated with month-to-month PM2.5 fluctuations. Adding weather controls alone (column 3) sharply increases F-statistic from 256 to 460. The increase in first-stage strength is primarily driven by temperature. Temperature independently affects PM2.5 through emissions and atmospheric chemistry and controlling for it removes variation in pollution unrelated to thermal inversion, sharpening the estimated relationship between the instrument and PM2.5. Column 4, the preferred specification, combines both sets of controls, with a result nearly identical to column 3, confirming that weather variables account for essentially all of the improvement in first-stage precision. The second-stage estimate is stable at approximately 0.038 across columns 2 through 4, indicating that the causal estimate itself is not sensitive to which controls are included, even as first-stage precision varies substantially across

specifications. Appendix E confirms this finding is robust to a more flexible temperature specification that includes monthly minimum and maximum temperatures.

The preferred specification produces a second-stage estimate of 0.038, indicating that a one-unit increase in PM2.5 is associated with a 3.8 percent increase in the cardiovascular mortality rate. At an average CVD mortality rate of 28.43 deaths per 100,000 residents per month, this corresponds to approximately 1.10 additional deaths per 100,000 per month[14]. For interpretability, I convert PM2.5 to its EPA AQI sub-index, the pollutant-specific component of the public air quality index.[15] A one-unit increase in the PM2.5 AQI sub-index is associated with a 1.35 percent increase in CVD mortality. As a check on instrument exogeneity, I regress the county unemployment rate on the instrument, controlling for weather and fixed effects. The small and insignificant estimate is consistent with thermal inversion not being driven by human activities [16]. Alternative instruments, including wind direction and PBLH, are examined in Appendix C. Appendix D shows that the instrument relationship is mainly contemporaneous, as lagged and lead inversion coefficients are small, and that the result holds regardless of which subset of the day's readings is used to build the instrument. It also shows that the instrument becomes much weaker when the panel is collapsed to the county-year level, which is consistent with thermal inversion being a short-run weather event and supports monthly aggregation as the appropriate level for this study (Appendix Table D3).

14. $28.43 \times [\exp(0.038) - 1] = 1.10$

15. The AQI is built from the same satellite PM2.5 data used throughout the paper to keep full county coverage. The publicly reported AQI is the maximum of six pollutant-specific sub-indices, with the reported value determined by whichever pollutant is worst that day; I compute the PM2.5 sub-index only for consistency with my paper. I use EPA's pre-2024 formula, which converts a PM2.5 concentration into a continuous value by linear interpolation within six bands: Good (0–50), Moderate (51–100), Unhealthy for Sensitive Groups (101–150), Unhealthy (151–200), Very Unhealthy (201–300), and Hazardous (301–500). These breakpoints applied during the 2010–2023 study period, before EPA's 2024 revision.

16. Coefficient is 0.0001 (0.0003), with a P-value of 0.712.

OLS estimates in the bottom panel are negative and downward-biased across all specifications. This likely reflects two sources of bias. First, although PM2.5 is measured from satellite-based data, measurement error in pollution exposure attenuates OLS estimates toward zero. Second, unobservable county characteristics that affect both pollution levels and cardiovascular health are not fully captured by the included controls, introducing omitted variable bias. The 2SLS estimates address both concerns by using exogenous variation in thermal inversion as an instrument, isolating the causal effect of PM2.5 on CVD mortality. Negative and downward-biased OLS estimates are common in the air pollution literature and have been documented across a range of health outcomes (Deschenes et al., 2020; Liu & Ao, 2021; X. Zhang et al., 2023).

Compared with other causal estimates of air pollution's impact on CVD mortality, the preferred estimate of 0.038 is broadly consistent with the causal literature. Deryugina et al. (2019) find significant causal effects of PM2.5 on cardiovascular mortality among elderly Medicare beneficiaries in the United States using wind direction as an instrument, though direct magnitude comparison is not straightforward due to differences in population and temporal aggregation. Zhu et al. (2024) also use an instrumental variable approach and find a positive and significant effect of PM2.5 on daily CVD mortality in China. Their setting, pollution levels, and time unit differ from mine, so the sizes of the two estimates are hard to compare directly.

V.d. Heterogeneity Analysis

I begin by examining why the population weighted estimate (0.013) is much smaller than the county average estimate (0.038)[17]. Table 8 splits the sample into urban and rural counties based on the 2013 NCHS Urban-Rural Classification Scheme for Counties[18]. PM2.5 raises CVD mortality in both groups, with a larger effect in rural counties. A one-unit increase in PM2.5 raises the CVD mortality rate by 4.1 percent in rural counties, compared to 2.4 percent in urban counties. This pattern is consistent with prior observational evidence of higher cardiovascular mortality risks from PM2.5 in rural areas (Garcia et al., 2016; Son et al., 2021), and may reflect more limited healthcare access and emergency response capacity in rural areas. Panel B shows that both groups remain significant with population weights. The rural estimate barely changes (0.044), while the urban estimate falls to 0.011. The smaller weighted estimate for the full sample therefore reflects the large weight on urban counties, where the effect is smaller, rather than a weaker effect overall.

Air pollution's health effects are not evenly distributed across the population; prior work finds that people of color experience a disproportionately higher burden of PM2.5 exposure (Tessum et al., 2021). Table 9 shows disparities by race, gender, age, and income. Among racial groups, PM2.5 has the highest impact on CVD mortality among the Black population, with an estimate of 3.1 percent close to the full sample baseline. While the White group also experiences an increase in CVD mortality rate caused by PM2.5, the estimate is notably smaller than that of the Black population. The impact on the Hispanic group is interestingly small and insignificant,

17. Weighting by the square root of county population, which gives large counties more weight than the unweighted specification but less than full population weights, gives an estimate of 0.023 (0.004), between the unweighted estimate of 0.038 and the population weighted estimate of 0.013.
18. Counties classified as large central metro, large fringe metro, medium metro, or small metro (codes 1–4) are coded as urban; counties classified as micropolitan or noncore (codes 5–6) are coded as rural.

as usually this group experiences disadvantaged conditions. This is consistent with the Hispanic mortality paradox that Hispanics in the U.S. often experience similar or better health outcomes than non-Hispanic Whites (Fernandez et al., 2023; Ruiz et al., 2013). Overall racial heterogeneity results are consistent with the literature (Di et al., 2017; Erqou et al., 2018), finding that the Black population faces a higher rate of CVD mortality caused by PM2.5 than other races and ethnicities. One possible explanation is unequal access to healthcare: long-standing residential segregation and disinvestment have left many predominantly Black neighborhoods with fewer primary care providers and specialists, and the facilities that do exist tend to be lower in quality (Bailey et al., 2017). This disparity is observed even in socioeconomic status adjusted models (B. Wang et al., 2020), suggesting race captures a vulnerability channel distinct from income. Heterogeneity analysis across genders shows slightly larger effects for women than for men. Gao et al. (2019) show that while CVD incidence is higher among men, women face higher mortality due to a cardiac event. Based on clinical evidence, they attribute this gender disparity to the underestimation of cardiovascular risk in women and less aggressive treatment. Therefore, this higher mortality response to PM2.5 among women in my work may reflect greater gender vulnerability and the gaps in diagnosis and treatment. Columns 7 to 10 show heterogeneity across age groups. While all age groups experience a rise in CVD mortality rate due to increased levels of PM2.5, the impact on the population above 85 years old as a vulnerable group to pollutant effects is higher. A one-unit increase in PM2.5 raises CVD mortality among those 85 and older by approximately 6.1%, about twice the baseline estimate. While both the 45-64 and 65-84 age groups experience nearly similar effects, the impact is 0.2 percentage points higher for 45-64 year-olds. Consistent with these results, Deryugina et al. (2019) also show a higher impact on the population aged 85 and older among Medicare beneficiaries. Columns 11 and 12 split

counties into low- and high-income groups based on the sample median of average county median household income[19]. Results show PM2.5 increases the mortality rate among low-income counties even higher than the baseline, while in high-income counties, there is no significant effect on the outcome. The larger air pollution impacts in low-income counties reveal the disproportionate effect of the air pollution burden on people from lower socioeconomic status. Income is only one part of poverty. Lower-income households also face more financial stress, less access to healthy food and safe places to exercise, and more trouble affording medication or getting to appointments, all of which raise cardiovascular risk on their own (Metkus, 2024). The larger air pollution impacts on Black and lower-income groups are consistent with Di et al. (2017), who studied air pollution effects on the Medicare population in the U.S.

Table 10 presents heterogeneity analysis by season, splitting the sample into cold season months (November to March) and warm season months (April to October). Both columns 2 and 3 show that PM2.5 significantly increases the CVD mortality rate in both seasons, but the effect is larger in the cold season (0.059) than in the warm season (0.018). This is expected, as thermal inversions are more common in winter and are driven by cold weather conditions[20] (Silcox et al., 2012). However, this does not imply a violation of the exclusion restriction. As stated in Section III, thermal inversions work through a pollution-trapping mechanism and have no direct effect on human health (Silcox et al., 2012). Moreover, weather controls already account for any direct effect of cold temperature on CVD mortality in all specifications, a claim further supported by Appendix E, which shows the estimate is stable when minimum and maximum monthly temperature are added as additional controls. These results are also in line with epidemiological

19. Counties are split into low- and high-income groups using the sample median, and income is also kept as a continuous control within each group. The split identifies which group a county falls into, while the continuous control accounts for the remaining income differences within that group.
20. High pressure systems, weak winter solar radiation, and radiational cooling

studies showing that PM2.5 effects on cardiovascular outcomes tend to be stronger in winter months (Hsu et al., 2017), though that study looks at hospitalization rather than mortality.

V.e. Robustness Checks

In this section, I first test the validity of the instrument using placebo outcomes. I then check whether the findings are robust to different causes of death, the functional form of the outcome, alternative measures of PM2.5, alternative pollutants, and the COVID-19 period.

One existing concern is that thermal inversion correlates with temperature and seasonal conditions that could directly affect CVD mortality independent of PM2.5. If this were the case, thermal inversion should also predict other outcomes sensitive to these conditions. I use homicide and accidental fall mortality rates per 100,000 residents as placebo outcomes. A separate literature finds that air pollution causally increases some types of violent crime, but Herrnstadt et al. (2021) find no significant effect on homicide when disaggregating crime by offense type. Moreover, Rahman et al. (2023) similarly finds that homicide mortality responds to temperature but not to air pollution exposure. For accidental falls, Guo et al. (2018) link long-term PM2.5 exposure to fall-related injury, which is non-fatal rather than mortality. I expect insignificant reduced-form estimates on these outcomes, suggesting that thermal inversion operates through PM2.5. Table 11 shows that thermal inversion has no statistically significant effect on either outcome, supporting the validity of the exclusion restriction[21].

Table 12 examines alternative mortality outcomes. PM2.5 exposure raises both total mortality and respiratory mortality but shows no significant effect on cancer mortality. The larger CVD effect fits with medical literature showing that cardiovascular disease is one of the

21. 0.00003 for homicide, $p = 0.503$; -0.0001 for accidental falls, $p = 0.113$.

major health consequences of PM2.5 exposure (Brook et al., 2010; Pope et al., 2004). Null finding on cancer mortality in line with Deryugina & Reif (2023) confirms that PM2.5 exposure produces immediate CVD effects, while cancer mortality reflects cumulative exposure over years.

Table 13 examines whether the main finding is sensitive to the functional form of the outcome. The baseline specification uses the natural logarithm of the CVD mortality rate, and I add one before taking the log to handle counties with zero deaths in a given month. This transformation allows coefficients to be interpreted as approximate percent changes in the mortality rate and makes the estimates less sensitive to outliers. Column 2 uses the inverse hyperbolic sine transformation of the CVD mortality rate, which also handles zero values without adding a constant and has a similar percent change interpretation to the baseline. The second stage estimate of 0.043 is almost identical to the baseline estimate of 0.038, confirming the result does not depend on this choice. Column 3 uses the CVD mortality rate as the outcome without any transformation. The estimate of 0.620 shows that a one-unit increase in PM2.5 raises the CVD mortality rate by 0.620 deaths per 100,000 residents per month. This larger magnitude is expected because the level outcome and the log outcome measure effects on different scales. The consistency of sign and significance across all three columns confirms that the main finding is not driven by the choice of outcome transformation.

Table 14 tests whether the main finding is sensitive to how PM2.5 is measured. Among alternative ways to measure PM2.5, column 2 maps satellite data to counties using inverse distance weighting rather than zonal statistics. Since both approaches use the same data source at the same spatial resolution, the two measures are highly correlated, and the results are nearly identical to the baseline. Column 3 replaces continuous PM2.5 with a binary indicator equal to

one when monthly PM2.5 exceeds the EPA annual standard of 12 μg/m³. The estimate remains positive and significant with a larger magnitude, which likely reflects that months above the threshold involve more severely polluted conditions. This confirms that the main finding is not an artifact of the linear specification and addresses potential nonlinearity in the pollution-mortality relationship. The baseline PM2.5 measure comes from satellite-based reanalysis data, so columns 4 and 5 compare this against ground-based data within the same sample of counties that have EPA monitoring stations to test the sensitivity of the findings to this change. Both columns cover 94,795 observations rather than the full 481,094 due to limited monitor coverage. The estimates of 0.017 and 0.014 are smaller than the baseline but remain positive and significant, and their similarity confirms that the two data sources give consistent results within the same counties.

In Table 15, I examine whether the main findings are sensitive to the choice of pollutants by replacing PM2.5 with alternative pollutants, standardizing all pollutants to mean zero and unit standard deviation to make coefficients comparable across pollutants with different units. All pollutants provide a positive and significant first stage, confirming the relevance condition with large F-statistics. Standardized PM2.5 and PM10 show similar estimates, as these two pollutants are highly correlated and mostly capture the same pollution sources[22]. Both CO and NO2 in columns 4 and 5 show positive and significant second-stage estimates, though smaller in magnitude than the PM2.5 coefficient in column 2. The greater detrimental impact of PM2.5 on CVD mortality rates confirms the higher impacts of PM pollutants on the human body, likely due to easier penetration. Schwartz et al. (2018) also find that PM2.5 has a robust effect on myocardial infarction hospitalizations, while the effects of CO and NO_2 are less consistent.

22. PM10 refers to the coarser fraction of particulate matter, up to 10 μm in diameter.

Unlike other pollutants, ozone in column 6 shows a negative first stage, consistent with findings in some other papers. Hicks et al. (2016) show that depending on the time of the day, inversion can increase or decrease ozone levels[23]. Janhäll et al. (2006) also document lower ozone levels during inversion periods. Given this negative first stage, thermal inversion is not a valid instrument for ozone, and the negative second stage estimate in column 6 reflects this identification failure rather than a true causal effect. This null ozone result is further consistent with Yu et al. (2024), who find no significant impact of ozone on CVD mortality.

This paper covers 2010-2023, including COVID-19 years, which increased cardiovascular mortality due to disrupted healthcare access (Woodruff et al., 2024) and changed pollution patterns through lockdown (Ghosal & Saha, 2021). Table 16A addresses concerns that COVID-19 may confound the estimated relationship between PM2.5 and CVD mortality. Column 2, which restricts the sample to the pre-pandemic period (2010 to 2019), shows an estimate identical to the baseline, confirming that the main result is not driven by COVID-19-related mortality patterns. Column 3 restricts the sample to the pandemic period (2020 to 2021) and has fewer observations than column 2, which reduces the sample size and widens the standard errors in this specification. The first stage remains strong, indicating that thermal inversion continues to predict PM2.5 during the pandemic, so the insignificant second stage is not due to instrument failure. Since cardiovascular mortality is defined using the underlying cause of death, COVID-19 deaths are not included in the outcome. To test whether the insignificant estimate in column 3 is driven by this exclusion, I combine CVD deaths and COVID-19 deaths into a single outcome in column 4. This change leads to a significant second-stage estimate close to the baseline, indicating that PM2.5 continued to raise mortality during the

23. Hicks et al. (2016) show that morning inversions tend to increase ozone while afternoon inversions reduce it.

pandemic but that some of these deaths were attributed to COVID-19 rather than cardiovascular disease. This is consistent with the medical literature cited in Austin et al. (2023) showing that PM2.5 damages the cardiovascular system, which in turn increases the risk of severe infectious disease progression such as COVID-19. Table 16B checks whether thermal inversion's exogeneity is compromised by COVID-19 severity. If that is the case, the bias would be larger in the hardest-hit counties. Columns 2 and 3 split the pandemic sample into counties above and below the median COVID-19 mortality[24]. Both estimates are insignificant with no clear difference between the two groups, which is consistent with the instrument's exogeneity holding during the pandemic period.

Table 17 tests for mortality harvesting to find whether PM2.5 affects individuals who were already at elevated risk of CVD death regardless of pollution exposure. If harvesting were a concern, current month pollution should reduce next month's mortality. Insignificant estimates in Column 3 using a one-month lag of PM2.5 show that last month's pollution did not reduce this month's mortality, confirming that harvesting is not a concern in this setting.

V.f. Policy Implications: Monetizing the Mortality Effect

Findings show detrimental impacts of PM2.5 exposure on the CVD mortality rate in the U.S., where CVD is the leading cause of death and imposes substantial costs on the country. To provide policymakers with a clear picture of these costs, I convert the main finding into a dollar value using the Environmental Protection Agency's value of statistical life (VSL), the standard tool used in federal regulatory cost-benefit analysis for mortality risk reductions.[25] For national

24. Columns 2 and 3 split the 2020-2021 sample at the median county-level average COVID-19 mortality rate of 12.4 per 100,000 (range: 0 to 183.7).

25. EPA's central VSL estimate is $7.4 million in 2006 dollars. I inflate this value once to 2023 dollars using the annual average Consumer Price Index for All Urban Consumers (CPI-U), published by the Bureau of Labor Statistics,

totals, I weight the preferred specification by county population, which gives the effect of 0.013 with standard error of 0.004 for the average resident. At a population-weighted national CVD mortality rate of 20.46 deaths per 100,000 residents per month, this corresponds to about 0.275 additional deaths per 100,000 residents per month ($20.46 \times [\exp(0.0133) - 1]$). Scaling this effect to the full U.S. population and over twelve months gives about 11,050 additional deaths in 2023 ($0.275 \times \frac{334{,}969{,}589}{100{,}000} \times 12$). Multiplying this figure by the 2023 VSL of $11.184 million gives an annual welfare cost of about $123.6 billion, with a 95 percent confidence interval of $45.5 billion to $202.3 billion. Because the effect and the mortality rate are held fixed and costs are expressed in constant 2023 dollars, costs in other years differ only through population, ranging from about $113.7 billion in 2010 to $123.6 billion in 2023 as reported in Table 18. For comparison, the EPA estimates that tightening the annual PM2.5 standard from 12 to 9 μg/m³ would avoid up to 4,500 premature deaths and yield net health benefits of up to $46 billion in 2032 (U.S. EPA, 2024)[26]. These estimates suggest that further reductions in PM2.5 could bring large health benefits.

VI. Conclusion

Cardiovascular disease is the leading cause of death in the United States, and air pollution is one of its known drivers. This study estimates the causal effect of PM2.5 on cardiovascular mortality at the U.S. national level using a county-month panel from 2010 to 2023. Most causal studies using thermal inversion as an instrument rely on a single data source and height selection without evaluating whether that construction is appropriate for their sample or how sensitive the results

yielding a VSL of $11.184 million. I apply this single value across all sample years rather than inflating separately for each year, so that reported welfare costs reflect real changes in mortality and population rather than changes in the price level.

26. https://www.epa.gov/system/files/documents/2024-02/pm-naaqs-overview.pdf

are to those choices. This paper fills that gap by systematically comparing construction choices across data products, height selections, aggregation methods, and spatial mapping approaches, and evaluating how each choice affects both stages of the 2SLS estimates.

This systematic evaluation shows that construction choices matter and that not all constructions are suitable for a national study. Fixed pressure levels at 1000 and 975 hPa, the approach used in most prior studies, are measured from sea level and fall below the surface in about 47 percent of grid cells and 30 percent of counties. In these high-elevation counties, the fixed pressure level instrument has no first stage, so it carries no information about the effect of PM2.5 there, and its national estimate comes mostly from low-elevation counties with a much weaker first stage. MERRA-2 model level data interpolated to 110 and 320 meters above local ground avoids this problem, since every grid point has valid temperature readings regardless of elevation, and gives a strong instrument in both high- and low-elevation counties. Among the construction choices evaluated, occurrence measures give stable estimates while strength measures give small or imprecise estimates, and the count-based occurrence instrument with zonal mapping at 110 and 320 meters provides a strong first stage and a high Conley breakpoint, indicating robustness to potential violations of the exclusion restriction. The importance of construction choices generalizes to any instrument built from raw geophysical or administrative data, since these all involve similar choices in how the measure is built. Testing the sensitivity of results to different choices should matter beyond this setting, although the terrain problem itself is specific to atmospheric data.

Using this preferred instrument shows that a one-unit increase in PM2.5 raises the cardiovascular mortality rate by 3.8 percent at the county-month level, corresponding to approximately 1.1 additional deaths per 100,000 residents per month. Results for alternative mortality outcomes show positive effects on total and respiratory mortality but no significant effect

on cancer mortality. The estimated effect on cardiovascular mortality is larger than the corresponding estimates for the alternative outcomes, confirming that CVD is one of the major health consequences of PM2.5 exposure. This finding is robust to alternative control sets, fixed effects structures, outcome transformations, alternative PM2.5 measures, alternative pollutants, and sample period restrictions. The similarity of estimates during the pre-pandemic period with the baseline confirms the result is not driven by COVID-19 related disruptions. One limitation of this work is that individual death records at the daily level are not available due to data access restrictions, which makes it difficult to study the effects at a finer time scale. A second limitation is that pollution exposure is assigned by county of residence, but commuting and relocation mean some residents' true exposure differs from their county average. This is unlikely to be correlated with thermal inversion timing, so it should attenuate rather than bias the estimate upward.

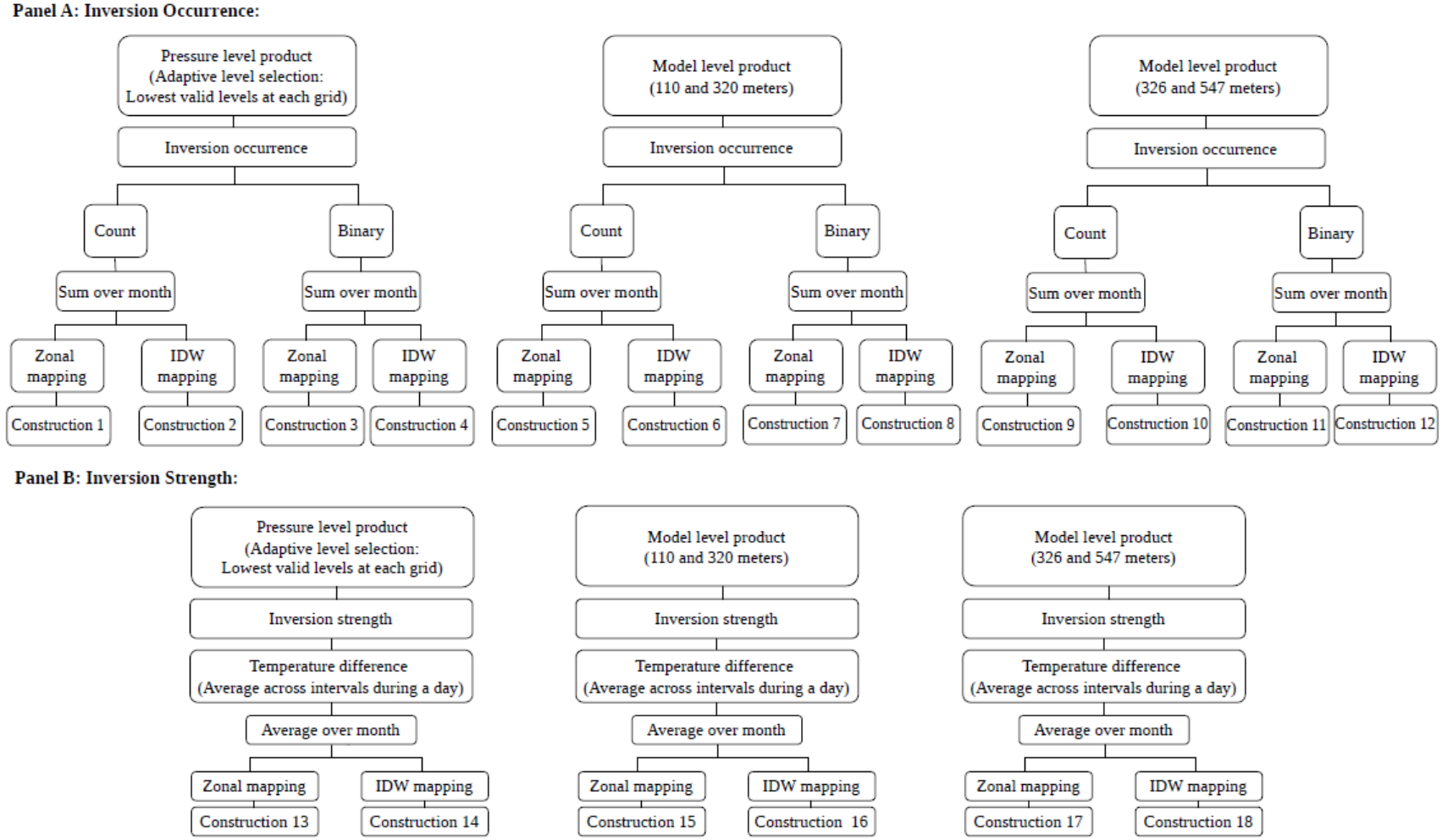
These results, along with the estimated annual cost of the findings based on EPA's value of a statistical life, suggest that further efforts to reduce PM2.5 levels can lower cardiovascular mortality and reduce the related costs to public health and the economy. This matters more in areas that exceed the EPA annual standard, which was 12 $\mu g/m^3$ during the sample period and was lowered to 9 $\mu g/m^3$ in 2024, since reducing pollution in these areas would have the largest impact on cardiovascular deaths. These findings suggest that people of color, low-income counties, the elderly, and rural populations bear a disproportionate share of PM2.5's cardiovascular mortality burden, and air quality regulations should account for this.

Future researchers can use the systematic instrument construction framework developed here to select the most appropriate construction for their sample and setting. Getting causal estimates right in this area matters because even a small effect at the individual level adds up to a large number of deaths across the country.

Acknowledgements

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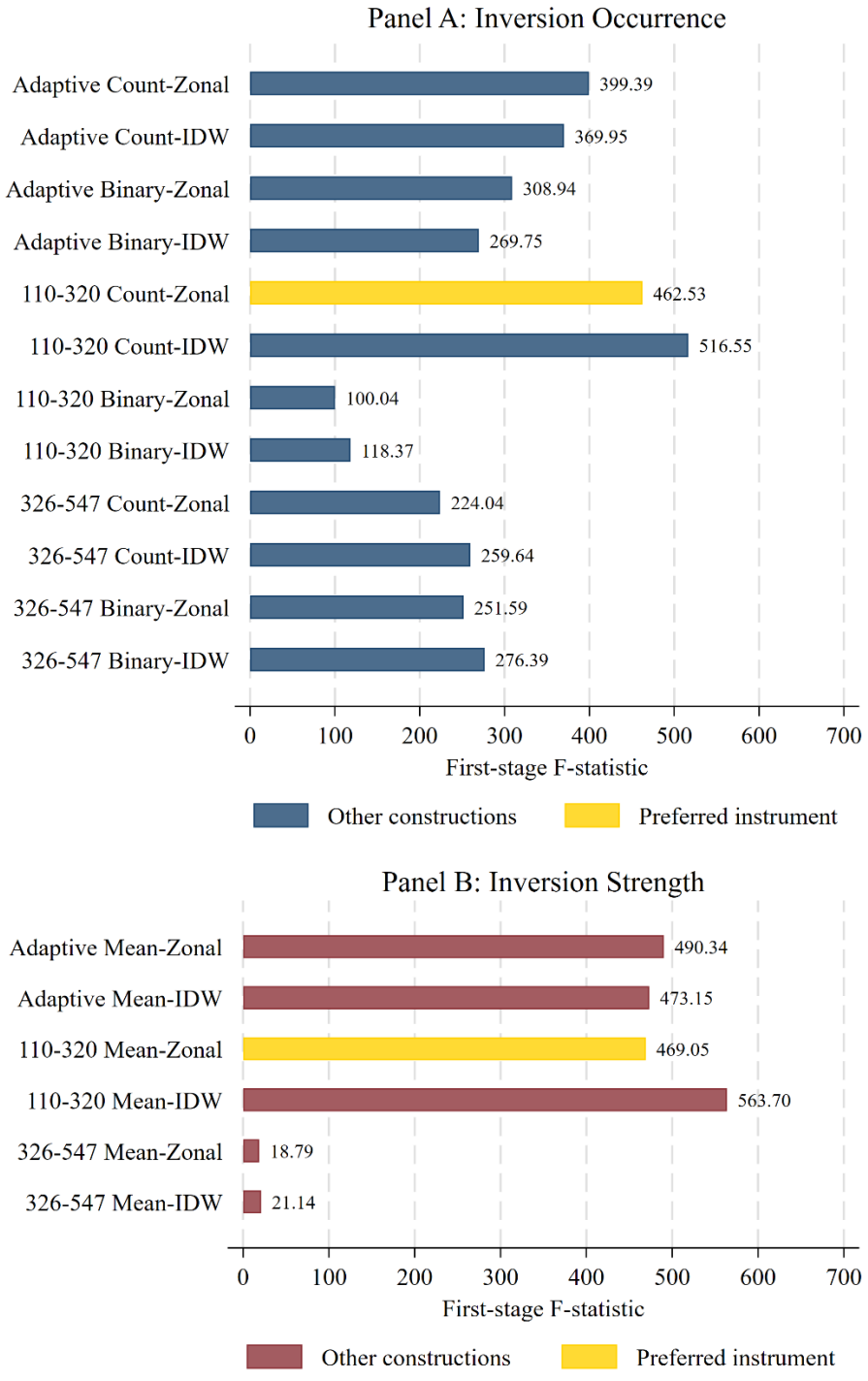
Figure 1. Construction of thermal inversion instruments



Source: Author's construction using MERRA-2 from NASA

Notes: This figure summarizes the construction of inversion occurrence and inversion strength instruments under three level selections shown above: adaptive pressure level selection, model level selection at 110 and 320 meters, and model level selection at 326 and 547 meters. Panel A shows the construction of the 12 inversion occurrence instruments and Panel B shows the construction of the 6 inversion strength instruments, numbered 1–18 as used in Tables 2 to 5. Zonal mapping takes the mean of grid cells overlapping the county polygon; IDW mapping takes a weighted average of grid points within 100 kilometers of the county centroid.

Figure 2. First-stage performance of occurrence and strength instrument constructions



Notes: Panel A reports first-stage F-statistics for the 12 inversion occurrence constructions and Panel B reports first-stage F-statistics for the 6 inversion strength constructions. The gold bar in each panel marks the preferred construction, selected using the conceptual criteria in Section IV.b. IDW mapping produces a higher F-statistic than the preferred zonal construction in both panels, but zonal mapping is preferred because it keeps the instrument tied to conditions inside the county.

Table 1. Descriptive Statistics (2010-2023)

Variable	Mean	SD	Min	Max
Panel A: Outcome of interest				
Cardiovascular disease mortality rate (per 100,000 individuals)	28.43	20.60	0	729.36
Ln (CVD mortality rate)	3.11	0.91	0	6.59
Panel B: Environmental variables:				
Air pollution; PM2.5 ($\mu g/m^3$)	10.11	4.56	1.18	211.60
Temperature (K)	289.18	10.38	254.02	315.75
Precipitation (mm)	2.80	1.83	0.00	24.14
Evaporation (mm)	-2.04	1.26	-7.76	0.01
Wind speed (m/s)	1.19	0.82	0.00	7.09
Panel C: County-month characteristics				
Unemployment rate	5.70	2.92	0.90	36.10
Median income	51,097.49	14,727.01	17,109	178,707
Share of population with at least high school diploma	0.76	0.05	0.21	0.999
Panel D: Placebo Outcomes				
Homicide rate (per 100,000 individuals)	0.363	1.155	0	7.575
Accidental fall mortality rate (per 100,000 individuals)	1.396	2.667	0	15.147
Number of observations	481,094	481,094	481,094	481,094

Source: NCHS, CDS, BLS, CB, and author's calculation

Table 2. Descriptive statistics for thermal inversion variables (2010-2023)

Variable	Mean	SD	Min	Max
Panel A: Inversion occurrence, adaptive pressure level selection				
Construction 1 (count, sum, zonal)	33.203	11.962	0	110
Construction 2 (count, sum, IDW)	33.245	11.797	0	112.448
Construction 3 (binary, sum, zonal)	19.271	5.357	0	31
Construction 4 (binary, sum, IDW)	19.310	5.247	0	31
Panel B: Inversion occurrence, model level selection (110 and 320 m)				
Construction 5 (count, sum, zonal)	77.106	24.498	0	220
Construction 6 (count, sum, IDW)	76.994	23.945	0	215.466
Construction 7 (binary, sum, zonal)	23.291	5.166	0	31
Construction 8 (binary, sum, IDW)	23.225	4.990	0	31
Panel C: Inversion occurrence, model level selection (326 and 547 m)				
Construction 9 (count, sum, zonal)	31.006	28.488	0	246
Construction 10 (count, sum, IDW)	30.985	28.190	0	234.431
Construction 11 (binary, sum, zonal)	11.173	6.821	0	31
Construction 12 (binary, sum, IDW)	11.153	6.701	0	31
Panel D: Inversion strength (°C), adaptive pressure level selection				
Construction 13 (mean, mean, zonal)	0.541	0.332	0	4.027
Construction 14 (mean, mean, IDW)	0.541	0.330	0	4.058
Panel E: Inversion strength (°C), model level selection (110 and 320 m)				
Construction 15 (mean, mean, zonal)	0.522	0.333	0	4.091
Construction 16 (mean, mean, IDW)	0.521	0.329	0	3.855
Panel F: Inversion strength (°C), model level selection (326 and 547 m)				
Construction 17 (mean, mean, zonal)	0.161	0.232	0	6.199
Construction 18 (mean, mean, IDW)	0.161	0.230	0	5.506

Source: NASA and author's calculations

Table 3. Plausibly Exogenous Breakpoints Across Instrument Constructions (2010–2023)

Construction Number	Level selection	Daily aggregation	Spatial mapping	Reduced form	Breakpoint
Panel A: Inversion occurrence					
Construction 1	Adaptive	Count	Zonal	0.0007	55%
Construction 2	Adaptive	Count	IDW	0.0008	60%
Construction 3	Adaptive	Binary	Zonal	0.0023	70%
Construction 4	Adaptive	Binary	IDW	0.0025	70%
Construction 5	Model level at 110 and 320 m	Count	Zonal	0.0005	65%
Construction 6	Model level at 110 and 320 m	Count	IDW	0.0005	65%
Construction 7	Model level at 110 and 320 m	Binary	Zonal	0.0023	65%
Construction 8	Model level at 110 and 320 m	Binary	IDW	0.0027	70%
Construction 9	Model level at 326 and 547 m	Count	Zonal	0.0004	55%
Construction 10	Model level at 326 and 547 m	Count	IDW	0.0004	55%
Construction 11	Model level at 326 and 547 m	Binary	Zonal	0.0014	45%
Construction 12	Model level at 326 and 547 m	Binary	IDW	0.0016	50%
Panel B: Inversion strength					
Construction 13	Adaptive	Mean ΔT	Zonal	0.0056	0%
Construction 14	Adaptive	Mean ΔT	IDW	0.0066	0%
Construction 15	Model level at 110 and 320 m	Mean ΔT	Zonal	0.0131	5%
Construction 16	Model level at 110 and 320 m	Mean ΔT	IDW	0.0120	0%
Construction 17	Model level at 326 and 547 m	Mean ΔT	Zonal	0.0242	10%
Construction 18	Model level at 326 and 547 m	Mean ΔT	IDW	0.0245	10%
Number of observations			481,094		

Notes: This table reports plausibly exogenous breakpoints following Conley et al. (2012) for all physically valid instrument constructions. The breakpoint is the share of the reduced form coefficient that the direct effect of the instrument on the outcome must reach before the second-stage estimate loses significance. A higher breakpoint indicates a more robust instrument. Panel A covers the 12 inversion occurrence instruments and Panel B covers the 6 inversion strength instruments. All specifications absorb county, year, and state×month fixed effects and include weather and socioeconomic controls.

Standard errors clustered at the county level.

Table 4. 2SLS estimates with inversion occurrence instruments

Level selection	Adaptive				110 and 320 meters				326 and 547 meters			
Grid day	Count	Count	Binary	Binary	Count	Count	Binary	Binary	Count	Count	Binary	Binary
Spatial mapping	Zonal	IDW	Zonal	IDW	Zonal	IDW	Zonal	IDW	Zonal	IDW	Zonal	IDW
Construction number	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Second stage	0.027***	0.032***	0.053***	0.058***	0.038***	0.033***	0.087***	0.079***	0.042***	0.041***	0.037***	0.036***
	(0.007)	(0.007)	(0.009)	(0.010)	(0.007)	(0.007)	(0.018)	(0.015)	(0.010)	(0.010)	(0.011)	(0.010)
First stage	0.026***	0.026***	0.045***	0.044***	0.014***	0.016***	0.027***	0.034***	0.010***	0.011***	0.039***	0.045***
	(0.001)	(0.001)	(0.003)	(0.003)	(0.001)	(0.001)	(0.003)	(0.003)	(0.001)	(0.001)	(0.002)	(0.003)
F-statistics	399.39	369.95	308.94	269.75	462.53	516.55	100.04	118.37	224.04	259.64	251.59	276.39
Number of observations	481,094	481,094	481,094	481,094	481,094	481,094	481,094	481,094	481,094	481,094	481,094	481,094

Notes: This table reports 2SLS estimates for the period 2010–2023. The dependent variable is the natural logarithm of the cardiovascular mortality rate per 100,000 residents at the county–month level. Constructions 1–4 use adaptive pressure level selection identifying the lowest valid pressure level at each grid point. Constructions 5–8 use MERRA-2 M2I3NVASM model level data interpolated to 110 and 320 meters above local ground. Constructions 9–12 use the same model level product interpolated to 326 and 547 meters above local ground. Both height pairs sit inside the mean planetary boundary layer height of 568 meters. All specifications absorb county, year, and state×month fixed effects and include weather controls (temperature, wind, evaporation, precipitation) and socioeconomic controls (unemployment rate, median income, high school diploma share).
Standard errors clustered at the county level in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5. 2SLS estimates with inversion strength instruments

Level selection	Adaptive pressure level		110 and 320 meters		326 and 547 meters	
Grid day	Average	Average	Average	Average	Average	Average
Grid month	Average	Average	Average	Average	Average	Average
Spatial mapping	Zonal	IDW	Zonal	IDW	Zonal	IDW
Construction number	(13)	(14)	(15)	(16)	(17)	(18)
Second stage	0.006	0.007	0.013*	0.011	0.058***	0.056**
	(0.007)	(0.008)	(0.008)	(0.007)	(0.027)	(0.027)
First stage	0.900***	0.889***	1.022***	1.128***	0.419***	0.439***
	(0.041)	(0.041)	(0.047)	(0.047)	(0.097)	(0.095)
F-statistics	490.34	473.15	469.05	563.70	18.79	21.14
Number of observations	481,094	481,094	481,094	481,094	481,094	481,094

Notes: This table reports 2SLS estimates for the period 2010–2023. The dependent variable is the natural logarithm of the cardiovascular mortality rate per 100,000 residents at the county–month level. Constructions 13–14 use adaptive pressure level selection. Constructions 15–16 use model level data at 110 and 320 meters above ground. Constructions 17–18 use the same model level product at 326 and 547 meters above local ground. Inversion strength assigns each grid point a daily value equal to the average temperature difference between upper and lower levels when an inversion is present, zero otherwise. All specifications absorb county, year, and state×month fixed effects and include weather controls (temperature, wind, evaporation, precipitation) and socioeconomic controls (unemployment rate, median income, high school diploma share).

Standard errors clustered at the county level in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6. Instrument reliability by county elevation

	(1)	(2)	(3)	(4)	(5)	(6)
Data product and level selection	Model level (110 and 320 m)			Pressure level (1000 and 975 hPa)		
Grid day	Count	Count	Count	Count	Count	Count
Spatial mapping	Zonal	Zonal	Zonal	Zonal	Zonal	Zonal
County coverage	All	High-elevation counties	Low-elevation counties	All	High-elevation counties	Low-elevation counties
Panel A: Unweighted						
Second stage	0.038***	0.041**	0.033***	0.028***	0.035	0.014*
	(0.007)	(0.017)	(0.006)	(0.011)	(0.024)	(0.008)
First stage	0.014***	0.013***	0.016***	0.021***	0.073	0.030***
	(0.001)	(0.001)	(0.001)	(0.002)	(0.049)	(0.002)
F-statistics	463	92	478	109.60	2.21	338.91
Panel B: Population-weighted						
Second stage	0.013***	0.028**	0.011**	0.007	0.012	0.008
	(0.004)	(0.014)	(0.005)	(0.007)	(0.013)	(0.006)
First stage	0.027***	0.020***	0.028***	0.050***	0.058**	0.055***
	(0.002)	(0.005)	(0.002)	(0.006)	(0.023)	(0.006)
F-statistics	137.74	15.82	163.44	74.92	6.15	84.83
Number of observations	481,094	139,184	341,910	481,094	139,184	341,910

Notes: This table reports 2SLS estimates for the period 2010–2023. The dependent variable is the natural logarithm of the cardiovascular mortality rate per 100,000 residents at the county-month level. Panel A gives each county equal weight and reports the effect for the average county. Panel B weights each county by its population and reports the effect for the average resident. Columns 1–3 use the preferred model-level instrument (count, zonal mapping, 110 and 320 meters above the surface). Columns 4–6 use the count-based, zonally-mapped instrument constructed from the 1000 and 975 hPa pressure levels. High-elevation counties are those where the area-weighted majority of overlapping grids have invalid temperature data at both the 1000 and 975 hPa levels (953 counties) and low-elevation counties are all remaining counties (2,154 counties). All specifications include weather and socioeconomic controls.

Standard errors clustered at the county level in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7. Effect of PM2.5 on CVD mortality rate

	(1)	(2)	(3)	(4)
Second stage	0.046***	0.042***	0.038***	0.038***
	(0.010)	(0.010)	(0.007)	(0.007)
First stage	0.009***	0.009***	0.014***	0.014***
	(0.001)	(0.001)	(0.001)	(0.001)
F-statistics	256.08	260.98	460.46	462.53
OLS	–0.005***	–0.005***	–0.006***	–0.005***
	(0.001)	(0.001)	(0.001)	(0.001)
County fixed effects	Y	Y	Y	Y
State×month fixed effects	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y
Environmental controls			Y	Y
County socioeconomic controls		Y		Y
Number of observations	481,094	481,094	481,094	481,094

Notes: This table reports 2SLS estimates for the period 2010–2023. The dependent variable is the natural logarithm of the cardiovascular mortality rate per 100,000 residents at the county month level. The instrument is count-based model level occurrence instrument at 110 and 320 meters above the ground using zonal mapping. All columns include county and state×month fixed effects. Columns 1–4 additionally include year fixed effects. The OLS row reports ordinary least squares estimates using the same controls and fixed effects as each column. Environmental controls include temperature, wind, evaporation, and precipitation. County socioeconomic controls include unemployment rate, median income, and high school diploma share.

Standard errors clustered at the county level in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8. Heterogeneity analysis by geography: Urban vs rural

	(1)	(2)	(3)
	Baseline	Urban	Rural
Panel A: Unweighted	0.038***	0.024***	0.041***
Second stage	(0.007)	(0.006)	(0.012)
First stage	0.014***	0.020***	0.012***
	(0.001)	(0.001)	(0.001)
F-statistics	462.53	329.44	289.07
Panel B: Population weighted			
Second stage	0.013***	0.011**	0.044***
	(0.004)	(0.004)	(0.009)
First stage	0.027***	0.029***	0.014***
	(0.002)	(0.003)	(0.001)
F-statistics	137.74	131.52	178.46
Number of observations	481,094	184,663	296,431

Notes: This table reports 2SLS estimates of the effect of PM2.5 on the natural logarithm of CVD mortality rate per 100,000 residents, split by urban and rural county classification using the 2013 NCHS Urban-Rural Classification Scheme (codes 1–4 classified as urban, codes 5–6 as rural). Panel A gives each county equal weight and reports the effect for the average county. Panel B weights each county by its population and reports the effect for the average resident. The instrument is count-based model level occurrence instrument at 110 and 320 meters above the ground using zonal mapping. All specifications absorb county, year, and state×month fixed effects. Weather controls are temperature, evaporation, precipitation, and wind speed. County socioeconomic controls include unemployment rate, median income, and high school diploma share.

Standard errors clustered at the county level in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 9. Heterogeneity analysis by race, gender, age, and income

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Baseline	White	Black	Hispanic	Female	Male	Age (Under 45)	Age (45-64)	Age (65-84)	Age (Above 85)	Low income	High income
Second stage	0.038***	0.010**	0.031***	0.010	0.047***	0.035***	0.015***	0.041***	0.039***	0.061***	0.066***	0.011
	(0.007)	(0.005)	(0.007)	(0.014)	(0.011)	(0.010)	(0.006)	(0.012)	(0.013)	(0.021)	(0.012)	(0.009)
Number of observations	481,094	438,570	428,139	436,953	438,570	438,570	438,570	480,596	438,569	438,482	240,628	240,466

Notes: This table reports 2SLS estimates of the effect of PM2.5 on subgroup-specific cardiovascular mortality rates at the county-month level. Each outcome is the natural logarithm of one plus the subgroup CVD mortality rate per 100,000 residents using subgroup-specific population as the denominator, except columns 11 and 12 which use total CVD mortality and split counties by whether average median household income falls below or above the sample median. Column 1 shows the preferred specification for reference. The instrument is the count-based model level occurrence instrument at 110 and 320 meters above the ground using zonal mapping. All specifications absorb county, year, and state×month fixed effects. Weather controls are temperature, evaporation, precipitation, and wind speed. County socioeconomic controls include unemployment rate, median income, and high school diploma share.

Standard errors clustered at the county level in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 10. Heterogeneity analysis by season

	(1)	(2)	(3)
	Baseline	Cold season	Warm season
Second stage	0.038***	0.059***	0.018***
	(0.007)	(0.011)	(0.006)
First stage	0.014***	0.013***	0.023***
	(0.001)	(0.001)	(0.001)
F-statistics	462.53	876.97	473.33
Number of observations	481,094	201,084	280,010

Notes: This table reports 2SLS estimates of the effect of PM2.5 on the natural logarithm of the cardiovascular mortality rate per 100,000 residents at the county month level. Column 1 shows the preferred specification for reference. Columns 2 and 3 split the sample into cold-season months and warm-season months, where cold seasons are defined as November through March and warm seasons are defined as April through October. The instrument is count-based model level occurrence instrument at 110 and 320 meters above the ground using zonal mapping. All specifications absorb county, year, and state×month fixed effects. Weather controls are temperature, evaporation, precipitation, and wind speed. County socioeconomic controls include unemployment rate, median income, and high school diploma share.
Standard errors clustered at the county level in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 11. Placebo test: Effect of thermal inversion on unrelated outcomes

	(1)	(2)	(3)
	Baseline	Homicide	Accidental falls
Reduced form	0.0005***	0.00003	–0.0001
	(0.0001)	(0.00004)	(0.0001)
Number of observations	481,094	481,094	481,094

Notes: This table reports reduced form estimates of the effect of thermal inversion on placebo outcomes unrelated to air pollution at the county-month level. Column 1 shows the preferred specification for reference. Column 2 uses the natural logarithm of one plus the homicide mortality rate per 100,000 residents and column 3 uses the natural logarithm of one plus the accidental fall mortality rate per 100,000 residents; this transformation retains county-months with zero deaths. Both rates are winsorized at the 99th percentile prior to analysis. The instrument is the count-based model-level occurrence instrument at 110 and 320 meters above the ground using zonal mapping. All specifications absorb county, year, and state×month fixed effects. Weather controls are temperature, evaporation, precipitation, and wind speed. County socioeconomic controls include unemployment rate, median income, and high school diploma share.
Standard errors clustered at the county level in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 12. Alternative mortality outcomes

	(1)	(2)	(3)	(4)
	Baseline CVD mortality	Total mortality	Respiratory mortality	Cancer mortality
Second stage	0.038***	0.017***	0.021***	–0.002
	(0.007)	(0.002)	(0.008)	(0.007)
Number of observations	481,094	481,094	481,094	481,094

Notes: This table reports 2SLS estimates of the effect of PM2.5 on the natural logarithm of total, respiratory, and cancer mortality rate per 100,000 residents at the county-month level. Column 1 shows the preferred specification for reference for CVD mortality rate per 100,000 residents. All specifications absorb county, year, and state×month fixed effects and include weather and socioeconomic controls.

Standard errors clustered at the county level in parentheses.

* p < 0.10, ** p < 0.05, *** p < 0.01

Table 13. Alternative outcome transformations

	(1)	(2)	(3)
	Baseline	CVD mortality rate (IHS transformation)	CVD mortality rate
Second stage	0.038***	0.043***	0.620***
	(0.007)	(0.008)	(0.142)
First stage	0.014***	0.014***	0.014***
	(0.001)	(0.001)	(0.001)
F-statistics	463	462.53	462.53
Number of observations	481,094	481,094	481,094

Notes: This table reports 2SLS estimates of the effect of PM2.5 on cardiovascular mortality using alternative outcome transformations at the county-month level. Column 1 reproduces the preferred specification using the natural logarithm of the cardiovascular mortality rate per 100,000 residents, defined as $\ln(1 + \text{death rate})$. Column 2 uses the inverse hyperbolic sine (IHS) transformation of the cardiovascular mortality rate per 100,000 residents. Column 3 uses the cardiovascular mortality rate in levels per 100,000 residents. The instrument is the count-based model-level occurrence instrument at 110 and 320 meters above the ground using zonal mapping. All specifications absorb county, year, and state×month fixed effects. Weather controls are temperature, evaporation, precipitation, and wind speed. County socioeconomic controls include unemployment rate, median income, and high school diploma share. All specifications include county, year, and state-by-month fixed effects.

Standard errors clustered at the county level in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 14. Alternative PM2.5 measurements

	(1)	(2)	(3)	(4)	(5)
	Baseline	PM2.5 (IDW)	PM2.5 (Threshold)	Satellite PM2.5 (EPA counties)	PM2.5 (EPA)
Second stage	0.038***	0.038***	0.337***	0.017**	0.014**
	(0.007)	(0.007)	(0.065)	(0.007)	(0.006)
First stage	0.014***	0.014***	0.002***	0.021***	0.025***
	(0.001)	(0.001)	(0.0001)	(0.002)	(0.002)
F-statistics	463	482.50	668.16	163.13	126.78
Number of observations	481,094	481,094	481,094	94,795	94,795

Notes: This table reports 2SLS estimates of the effect of PM2.5 on the natural logarithm of the cardiovascular mortality rate per 100,000 residents at the county-month level. Column 1 shows the preferred specification for reference. Column 2 uses PM2.5 mapped by IDW rather than zonal statistics. Column 3 replaces PM2.5 with a binary indicator equal to one when PM2.5 exceeds the EPA annual standard of 12 $\mu g/m^3$. Column 4 restricts the sample to counties with EPA monitoring stations while retaining satellite-based PM2.5. Column 5 uses PM2.5 measured directly from EPA monitoring stations. All specifications absorb county, year, and state×month fixed effects and include weather and socioeconomic controls.

Table 15. Alternative pollutants

	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline	PM2.5 (std)	PM10 (std)	CO (std)	NO_2 (std)	Ozone (std)
Second stage	0.038***	0.171***	0.166***	0.105***	0.114***	–0.286***
	(0.007)	(0.033)	(0.032)	(0.020)	(0.022)	(0.058)
First stage	0.014***	0.003***	0.003***	0.005***	0.005***	–0.002***
	(0.001)	(0.0001)	(0.0001)	(0.0001)	(0.0002)	(0.0002)
F-statistics	462.53	462.53	442.78	1004.33	583.41	132.11
Number of observations	481,094	481,094	481,094	481,094	481,094	481,094

Notes: Column 1 shows the preferred specification for reference. All pollutants in columns 2 to 6 are standardized to mean zero and unit standard deviation. All columns include county, year, state×month. The instrument is count-based model level occurrence instrument at 110 and 320 meters above the ground using zonal mapping. All specifications absorb county, year, and state×month fixed effects. Weather controls are temperature, evaporation, precipitation, and wind speed. County socioeconomic controls include unemployment rate, median income, and high school diploma share.
Standard errors clustered at the county level in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 16A. Sensitivity to sample period and cause-of-death reclassification during COVID-19

	(1)	(2)	(3)	(4)
	Baseline (2010-2023)	Pre-COVID (2010-2019)	COVID years (2020-2021)	COVID years (2020-2021)
Outcome	CVD mortality	CVD mortality	CVD mortality	CVD and Covid mortality
Second stage	0.038***	0.038***	0.007	0.038*
	(0.007)	(0.009)	(0.020)	(0.020)
First stage	0.014***	0.014***	0.012***	0.012***
	(0.001)	(0.001)	(0.002)	(0.002)
F-statistics	463	402.25	56.43	56.43
Number of observations	481,094	335,992	72,536	72,536

Notes: This table reports 2SLS estimates. The dependent variable in columns 1-3 is the natural logarithm of the cardiovascular mortality rate per 100,000 residents. Column 4 uses a combined outcome summing cardiovascular and COVID-19 deaths before converting to a rate per 100,000 and taking the natural logarithm, following the same convention as the main outcome. Column 2 restricts the sample to 2010-2019, before the COVID-19 pandemic. Columns 3 and 4 restrict the sample to 2020-2021, the acute pandemic period. The instrument is the count-based model-level occurrence instrument at 110 and 320 meters above the surface using zonal mapping. All specifications absorb county, year, and state×month fixed effects and include weather and socioeconomic controls. Standard errors clustered at the county level in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 16B. Sensitivity to COVID-19 severity

	(1)	(2)	(3)
	Baseline (2010-2023)	High COVID counties (2020-2021)	Low COVID counties (2020-2021)
Second stage	0.038***	–0.011	0.022
	(0.007)	(0.025)	(0.024)
First stage	0.014***	0.018***	0.011***
	(0.001)	(0.001)	(0.002)
F-statistics	463	178.85	20.14
Number of observations	481,094	35,979	36,557

Notes: This table reports 2SLS estimates for the period 2020-2021. The dependent variable is the natural logarithm of the cardiovascular mortality rate per 100,000 residents. Column 2 restricts the sample to counties with above-median COVID-19 mortality during 2020-2021; column 3 restricts to counties at or below the median. This test addresses the concern that thermal inversion's exogeneity could be compromised by COVID-19 severity; the absence of a systematic difference between the two groups is consistent with the instrument's exogeneity holding during the pandemic period. The instrument is the count-based model-level occurrence instrument at 110 and 320 meters above the surface using zonal mapping. All specifications absorb county, year, and state×month fixed effects and include weather and socioeconomic controls.
Standard errors clustered at the county level in parentheses.
* p < 0.10, ** p < 0.05, *** p < 0.01

Table 17. Mortality harvesting test: Distributed lag estimates

	(1)	(2)	(3)
	Baseline	PM2.5	PM2.5 (One month lag)
Second stage	0.038***	0.022***	0.011
	(0.007)	(0.006)	(0.008)
Number of observations	481,094	462,709	462,709

Notes: This table reports 2SLS estimates of the effect of PM2.5 on the natural logarithm of the cardiovascular mortality rate per 100,000 residents at the county-month level. Column 1 shows the preferred specification for reference. Columns 2 and 3 instrument current and lagged PM2.5 with current and lagged thermal inversion to test for mortality harvesting. A negative and significant one month lag coefficient would indicate that pollution displaces imminent deaths rather than causing genuine mortality. All specifications absorb county, year, and state×month fixed effects and include weather and socioeconomic controls. Standard errors clustered at the county level in parentheses.

Standard errors clustered at the county level in parentheses.

* p < 0.10, ** p < 0.05, *** p < 0.01

Table 18. Monetized cost of a 1 μg/m³ increase in PM2.5 by year

	(1)	(2)
Year	Extra deaths	Welfare cost (billion dollars)
2010	10,164	113.7
2011	10,248	114.6
2012	10,336	115.6
2013	10,411	116.4
2014	10,510	117.5
2015	10,590	118.4
2016	10,666	119.3
2017	10,734	120.1
2018	10,791	120.7
2019	10,843	121.3
2020	10,937	122.3
2021	10,954	122.5
2022	10,996	123.0
2023	11,050	123.6

Notes: This table reports the estimated additional annual CVD deaths and monetized annual welfare cost of a sustained nationwide 1 μg/m³ increase in PM2.5, by year. Estimates use the population-weighted 2SLS coefficient of 0.0133 and the population-weighted national CVD mortality rate of 20.46 deaths per 100,000 residents per month. Welfare cost uses a fixed VSL of $11.184 million (2023 dollars), applied uniformly across years.

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Online Appendix:

Appendix A: Comparative Overview of Literature

Table A1. Comparative overview of air pollution health effects by method, instrument, and outcome

Study	Country Years	Methodology	Outcome	Pollutant Level	Coefficient
This paper	USA 2010-2023	2SLS IV: thermal inversion	CVD mortality rate	PM2.5 County Month	• 0.038
Danesh Yazdi et al. (2022)	USA 2001-2016	DiD	CVD hospital admission (per 100,000 individuals)	PM2.5 Zip-code Annual	• 47.71
Fan et al. (2023)	China 2004-2012	DiD 2SLS IV: SO_2 reduction policy	CVD mortality rate (per 100,000 individuals)	SO_2 City Annual	• DiD: -20.973 • IV:1.596
Mazidi & Speakman (2018)	USA 2003-2011	Cross-sectional linear regression	CVD mortality	PM2.5 County Annual	• 1.2
Hart et al. (2015)	USA 1989-2006	Spatio-temporal prediction model (Cox proportional hazards models)	Risk of CVD incident among women from the Nurses' Health Study cohort.	PM2.5 Individual Month	• 1.02

Note: Table compares the main coefficients from this paper with estimates from selected studies that either examine the same health outcomes using different methods or apply the same identification strategy (2SLS) but study different health outcomes. The included studies all examine health-related effects of air pollution, with a primary focus on PM2.5. In cases where a study analyzes multiple pollutants, only the coefficient for PM2.5 is reported for comparability. A few included studies examine other pollutants (such as CO) and are noted accordingly.

Table A1 (continued). Comparative overview of air pollution health effects by method, instrument, and outcome

Study	Country Years	Methodology	Outcome	Pollutant Level	Coefficient
You & Chou (2025)	USA 2001-2012	2SLS IV: thermal inversion	Mentally unwell days Asthma	PM2.5 County Week	• 0.11 • 0.16
Li and Zhang (2024)	China 1991-2000	2SLS IV: thermal inversion	Low birth weights (LBW)	PM2.5 Individual Month	• 0.400
Zhang et al. (2023)	China 2010-2018	2SLS IV: thermal inversion	Obesity	PM2.5 Individual County Annual	• 0.0286
Deschenes et al. (2020)	China 1991-2000	2SLS IV: thermal inversion	Body weight Obesity	PM2.5 County Annual	• 0.0082 (overweight) • 0.0021 (obesity)
Zhang et al. (2023)	China 2013-2015	2SLS IV: thermal inversion	Health spending	PM2.5 Individual County Month	• 2.361
S. Wang et al. (2023)	China 1991-2015	2SLS IV: thermal inversion	Healthcare expenditures	PM2.5 Individual County Annual	• 0.0510
Liu and Ao (2021)	Taiwan 2006-2012	2SLS IV: thermal inversion	Respiratory health expenditures	PM2.5 Individual Township Day	• 0.39
Arceo et al. (2016)	Mexico 1997-2006	2SLS IV: thermal inversion	Infant mortality (per 100,000 births)	PM10 Municipality Week	• 0.23

Note: Table compares the main coefficients from this paper with estimates from selected studies that either examine the same health outcomes using different methods or apply the same identification strategy (2SLS) but study different health outcomes. The included studies all examine health-related effects of air pollution, with a primary focus on PM2.5. In cases where a study analyzes multiple pollutants, only the coefficient for PM2.5 is reported for comparability. A few included studies examine other pollutants (such as CO) and are noted accordingly.

Table A1 (continued). Comparative overview of air pollution health effects by method, instrument, and outcome

Study	Country Years	Methodology	Outcome	Pollutant Level	Coefficient
Hicks et al. (2016)	USA 1980-2004	IV with LIML (limited-information maximum likelihood)	Cardiorespiratory mortality	CO County Month	• 3.17
Austin et al. (2023)	USA 2020	2SLS IV: wind direction	COVID-19 morbidity and mortality	PM2.5 County Day	• 0.009
Deryugina et al. (2019)	USA 1999-2013	2SLS IV: wind direction	Elderly mortality	PM2.5 County Day	• CVD mortality: 0.255
Gu et al. (2020)	China 2014	2SLS IV: maximum wind speed	Mental health	PM2.5 Individual City/County Annual	• 0.012

Note: Table compares the main coefficients from this paper with estimates from selected studies that either examine the same health outcomes using different methods or apply the same identification strategy (2SLS) but study different health outcomes. The included studies all examine health-related effects of air pollution, with a primary focus on PM2.5. In cases where a study analyzes multiple pollutants, only the coefficient for PM2.5 is reported for comparability. A few included studies examine other pollutants (such as CO) and are noted accordingly.

Appendix B: Fixed Pressure Levels Constructions

Most prior studies build thermal inversions from the MERRA-2 pressure levels at 1000 and 975 hPa, about 110 and 320 meters above sea level. Since these levels are measured from sea level, they fall below the surface in about 47 percent of grid cells and 30 percent of counties. Readings where either level is below the surface are counted as no inversion. Table B1 compares each fixed pressure level construction with the model level construction that uses the same aggregation and mapping choices at the same nominal heights above the local surface. Across all six pairs, the two approaches give estimates of the same sign, but the model level constructions have much stronger first stages. For count with zonal mapping, the F-statistic is 463 at model levels and 110 at fixed pressure levels. Strength instruments give small estimates under both approaches. Section V.b shows that the fixed pressure level instrument has no significant first stage in high-elevation counties.

Table B1. Model level and fixed pressure level constructions with the same aggregation and mapping choices

	Inversion occurrence				Inversion Strength	
Grid day	Count	Count	Binary	Binary	Average	Average
Grid month	Sum	Sum	Sum	Sum	Average	Average
Spatial mapping	Zonal	IDW	Zonal	IDW	Zonal	IDW
Panel A: Model level (110 and 320 meters above ground)						
Construction number	(5)	(6)	(7)	(8)	(15)	(16)
Second stage	0.038***	0.033***	0.087***	0.079***	0.013*	0.011
	(0.007)	(0.007)	(0.018)	(0.015)	(0.008)	(0.007)
First stage	0.014***	0.016***	0.027***	0.034***	1.022***	1.128***
	(0.001)	(0.001)	(0.003)	(0.003)	(0.047)	(0.047)
F-statistics	462.53	516.55	100.04	118.37	469.05	563.70
Panel B: Fixed pressure level (1000 and 975 hPa)						
Second stage	0.028**	0.037***	0.046**	0.058***	0.006	0.012
	(0.011)	(0.011)	(0.019)	(0.020)	(0.013)	(0.013)
First stage	0.021***	0.021***	0.027***	0.027***	0.681***	0.708***
	(0.002)	(0.002)	(0.004)	(0.004)	(0.074)	(0.068)
F-statistics	109.60	116.40	45.17	42.78	85.39	107.06
Number of observations	481,094	481,094	481,094	481,094	481,094	481,094

Notes: This table reports 2SLS estimates for the period 2010 to 2023. The dependent variable is the natural logarithm of the cardiovascular mortality rate per 100,000 residents at the county and month level. Panel A shows estimates using inversion construction built based on model level data at 110 and 320 meters. Panel B builds the same six measures from the fixed 1000 and 975 hPa pressure levels in MERRA-2. When the 1000 hPa level is below the ground surface, the reading is counted as no inversion. The two panels use the same daily aggregation, monthly aggregation and spatial mapping; only the levels used to measure the inversion differ. All specifications include county, year, and state by month fixed effects, along with weather and socioeconomic controls.
Standard errors clustered at the county level in parentheses.
* p<0.10, ** p<0.05, *** p<0.01.

Appendix C: Alternative Instruments

Wind direction is another commonly used instrument for air pollution, constructed using upwind industrial source exposure at the daily level (Austin et al., 2023; Deryugina et al., 2019). Applying wind direction in this study requires adaptation because the analysis is conducted at the county-month level. I construct county-month wind direction components by averaging the daily east-west and north-south wind vector components over each month. These are represented by sine and cosine components to account for the circular nature of directional data, where the sine component captures the east-west signal and the cosine component captures the north-south signal.

Unlike thermal inversion, for which the pollution-trapping mechanism implies a positive first-stage relationship with PM2.5, no theoretically predicted sign exists for the wind direction components because the relationship between wind direction and pollution depends on local geography and the spatial distribution of emission sources. As shown in Table C1, the first-stage F-statistics in columns 2 through 5 range from 25.6 to 68.9, confirming that wind direction is a relevant predictor of PM2.5 in these specifications. Instrument relevance, however, is distinct from the stability of the resulting second-stage estimate.

Across columns 2 through 5, the second-stage coefficient is consistently positive but varies substantially in magnitude and statistical significance depending on the included controls. The estimate is small and statistically insignificant in columns 2 and 3 but becomes significant once environmental controls are introduced in column 4 and increases to 0.059 in column 5, which includes the full set of controls comparable to the preferred thermal inversion specification. This estimate implies that a one-unit increase in PM2.5 increases the cardiovascular mortality rate by 5.9 percent, compared with 3.8 percent under the preferred

thermal inversion instrument. This is likely because wind direction is naturally a daily measure and averaging it to the monthly level smooths away most of the variation that makes it useful as an instrument. What's left is a slower, seasonal pattern, which is more sensitive to whatever else is included in the model. By contrast, the preferred thermal inversion instrument in Table 7 yields a stable second-stage estimate of approximately 0.038 regardless of whether controls are included. Column 6 uses thermal inversion jointly with the wind direction components as excluded instruments. The Hansen J overidentification test does not reject the null hypothesis that the instruments identify the same causal parameter ($p = 0.742$), and the resulting second-stage estimate of 0.039 is nearly identical to the baseline estimate obtained using thermal inversion alone. Although the combined specification produces a larger first-stage F-statistic than the wind-direction-only specifications, this improvement is primarily attributable to the strong predictive power of the thermal inversion instrument.

Another possible instrument is PBLH, which is correlated with thermal inversion. As described in the data section, a higher boundary layer allows pollutants to disperse, reducing ground level concentrations, while thermal inversion occurs within this layer by capping vertical mixing. Table C2 uses PBLH instead of thermal inversion. The first stage is negative and significant, consistent with the expectation that a higher boundary layer reduces pollution concentrations, and F-statistics are larger than those in Table 7. However, the second stage is negligible and insignificant across all columns. A strong first stage confirms relevance but does not guarantee a valid instrument. PBLH captures smooth seasonal and geographic variation in atmospheric mixing depth that may correlate with CVD mortality through channels unrelated to pollution, making the exclusion restriction difficult to defend. By contrast, thermal inversion

produces consistent and significant second stage estimates across all specifications in Table 7, supporting its use as the preferred instrument.

Table C1. Effect of PM2.5 on CVD mortality rate: Using wind direction as the instrument

	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline IV: thermal inversion	IV: wind direction; decomposed to sin and cos				IV: thermal inversion and wind direction
Second stage	0.038***	0.006	0.020	0.071**	0.059**	0.039***
	(0.007)	(0.015)	(0.016)	(0.030)	(0.029)	(0.007)
First stage (thermal inversion)	0.014***					0.014***
	(0.001)					(0.001)
First stage (sin)		–0.156***	–0.153***	0.033*	0.033*	0.012
		(0.014)	(0.014)	(0.017)	(0.018)	(0.018)
First stage (cos)		–0.088***	–0.090***	–0.112***	–0.113***	–0.119***
		(0.017)	(0.017)	(0.017)	(0.017)	(0.016)
F-statistics	463	68.93	62.63	25.58	26.00	154.67
Hansen J (P-value)	-	0.016	0.819	0.566	0.907	0.742
County fixed effects	Y	Y	Y	Y	Y	Y
State×month fixed effects	Y	Y	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y	Y
Environmental controls	Y			Y	Y	Y
County socioeconomic controls	Y		Y		Y	Y
Number of observations	481,094	481,094	481,094	481,094	481,094	481,094

Notes: This table reports 2SLS estimates for the period 2010-2023. The dependent variable is the natural logarithm of the cardiovascular mortality rate per 100,000 residents at the county-month level. Column 1 reproduces the preferred specification from Table 7 using the count-based thermal inversion occurrence instrument at 110 and 320 meters above local ground with zonal mapping. Columns 2-5 instrument PM2.5 with monthly wind direction decomposed into sine and cosine components to account for the circular nature of directional data, varying fixed effects and controls across columns. Column 2 includes no controls beyond fixed effects, column 3 adds socioeconomic controls, column 4 adds environmental controls, and column 5 includes all controls. Column 6 uses both thermal inversion and wind direction components simultaneously as excluded instruments. The Hansen J overidentification test evaluates the null hypothesis that all instruments are valid and identify the same parameter. A significant J statistic indicates that the instruments are not jointly valid or are identifying different parameters. All columns include county and state×month fixed effects. Environmental controls include temperature, wind, evaporation, and precipitation. County socioeconomic controls include unemployment rate, median income, and high school diploma share.

Table C2. Effect of PM2.5 on CVD mortality rate: Using PBLH as the instrument

	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline IV: thermal inversion	IV: PBLH				IV: thermal inversion and PBLH
Second stage	0.038*** (0.007)	0.006 (0.005)	0.002 (0.005)	0.006 (0.005)	0.00002 (0.005)	0.009* (0.005)
First stage (thermal inversion)	0.014*** (0.001)					0.007*** (0.001)
First stage (PBLH)		–0.004*** (0.0001)	–0.004*** (0.0002)	–0.004*** (0.0001)	–0.004*** (0.0002)	–0.003*** (0.0002)
F-statistics	463	678.77	486.23	679.92	488.93	297.11
Hansen J (P-value)	-	-	-	-	-	0.000
County fixed effects	Y	Y	Y	Y	Y	Y
State×month fixed effects	Y	Y	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y	Y
Environmental controls	Y			Y	Y	Y
County socioeconomic controls	Y		Y		Y	Y
Number of observations	481,094	481,094	481,094	481,094	481,094	481,094

Notes: This table reports 2SLS estimates for the period 2010-2023. The dependent variable is the natural logarithm of the cardiovascular mortality rate per 100,000 residents at the county-month level. The baseline column reproduces the preferred specification from Table 7 using the count-based thermal inversion occurrence instrument at 110 and 320 meters above local ground with zonal mapping. Columns 2-5 instrument PM2.5 with average planetary boundary layer height (PBLH), varying fixed effects and controls across columns similarly to Table C1. Column 6 uses both thermal inversion and PBLH simultaneously as excluded instruments. All columns include county and state×month fixed effects. Environmental controls include temperature, wind, evaporation, and precipitation. County socioeconomic controls include unemployment rate, median income, and high school diploma share.

Standard errors clustered at the county level in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Appendix D: Instrument Timing and Temporal Aggregation

To examine whether the instrument relationship is contemporaneous, Table D1 reports first stage regressions of PM2.5 on contemporaneous, lagged, and lead thermal inversion separately. The contemporaneous coefficient of 0.014 confirms the expected positive relationship between inversion and pollution. The lagged inversion coefficient is negative and small, about 6 percent of the contemporaneous coefficient, and significant only at the 10 percent level, and the lead inversion coefficient is small and insignificant, confirming that the instrument relationship is mainly contemporaneous.

To test whether the result depends on which subset of the day's readings is used, Table D2 reports second stage estimates using inversion counts built from two non-overlapping subsets of model level data available for eight daily 3-hourly UTC timestamps. Column 2 uses the four readings between 12:00 and 21:00 UTC, and column 3 uses the remaining four readings between 00:00 and 09:00 UTC. Both windows produce a strong first stage and a positive, significant second stage estimate, with a larger estimate in column 3. This confirms the main result is not driven by inversions occurring in a specific subset of the day's readings.

Table D3 reports the sensitivity of the results to the level of temporal aggregation. Column 1 shows the preferred monthly specification. Columns 2 and 3 collapse the panel to the county-year level. With state×year fixed effects in column 2, the instrument is still relevant but much weaker than at the monthly level, with a first-stage F-statistic of 17.64 compared with 463. With an F-statistic this low, the column 2 estimate is imprecise, and I do not interpret its size. With year fixed effects in column 3, the first stage is not significant and the second-stage estimate is not informative. Identification at the annual level depends on controlling for regional

time trends, and even then the instrument is far weaker than at the monthly level. This is consistent with thermal inversion acting as a short-run weather event.

Table D1. Lead and lag sensitivity of the first stage

	(1)	(2)	(3)
	Baseline	Lagged inversion	Lead inversion
First stage	0.014***	–0.0009*	0.0009
	(0.001)	(0.0005)	(0.0006)
Number of observations	481,094	462,709	462,709

Notes: This table reports first stage coefficients from separate regressions of PM2.5 on contemporaneous, lagged, and lead thermal inversion. Column 1 shows the baseline first stage using contemporaneous inversion. Column 2 replaces the instrument with its one month lag and column 3 replaces it with its one month lead. All specifications absorb county, year, and state×month fixed effects and include weather and socioeconomic controls.
Standard errors clustered at the county level in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table D2. Instrument construction using different daily reading windows

	(1)	(2)	(3)
	Baseline	12:00–21:00 UTC	00:00–09:00 UTC
Second stage	0.038***	0.023***	0.050***
	(0.007)	(0.007)	(0.009)
First stage	0.014***	0.035***	0.017***
	(0.001)	(0.002)	(0.001)
F-statistics	463	549.73	329.89
Number of observations	481,094	481,094	481,094

Notes: Dependent variable is natural logarithm of CVD mortality rate. Column 2 instruments PM2.5 using an inversion count built only from readings between 12:00 and 21:00 UTC model level data available every three hours. Column 3 uses only readings between 00:00 and 09:00 UTC. Both use the same zonal, count based construction as the preferred instrument. All specifications absorb county, year, and state×month fixed effects and include weather and socioeconomic controls.

Standard errors clustered at the county level in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table D3. Sensitivity to the level of temporal aggregation

	(1)	(2)	(3)
Second stage	0.038***	3.76***	–39.44
	(0.007)	(0.91)	(30.13)
First stage	0.014***	0.0005***	–0.0001
	(0.001)	(0.0001)	(0.0001)
F-statistics	463	17.64	1.71
County fixed effects	Y	Y	Y
State×month fixed effects	Y		
State×year fixed effect		Y	
Year fixed effects	Y		Y
Environmental controls	Y	Y	Y
County socioeconomic controls	Y	Y	Y
Number of observations	481,094	43,105	43,119

Notes: Column 1 reports the preferred monthly specification. Columns 2 and 3 collapse the panel to the county-year level, summing death counts and the inversion instrument and averaging PM2.5 and weather controls within each county-year. Column 2 drops 14 singleton observations from the inclusion of state×year fixed effects. All columns include environmental and socioeconomic controls.
Standard errors clustered at the county level are in parentheses.
*** p<0.01, ** p<0.05, * p<0.10.

Appendix E: Weather Control Sensitivity

The preferred specification in Table 7 includes all socioeconomic controls and environmental controls, including temperature, evaporation, precipitation, and wind speed. To address the concern that extreme temperature, either cold or hot, could be driving the results, I include the county-month minimum and maximum temperature as additional control variables. Table E shows the result is in line with the main finding of 0.038 and including minimum and maximum temperature results in a coefficient of 0.041, indicating a one-unit increase in PM2.5 raises the CVD mortality rate by 4.1 percent.

Table E. Sensitivity to temperature control specification

	(1)	(2)
	Baseline	Including min/max temperature
Second stage	0.038***	0.041***
	(0.007)	(0.007)
First stage	0.014***	0.014***
	(0.001)	(0.001)
F-statistics	463	441.04
Number of observations	481,094	481,094

Notes: This table reports 2SLS estimates of the effect of PM2.5 on the natural logarithm of the cardiovascular mortality rate per 100,000 residents at the county-month level. Column 1 shows the preferred specification from Table 7 for reference. Column 2 adds county-month minimum and maximum temperature to the set of weather controls. All specifications absorb county, year, and state×month fixed effects and include weather and socioeconomic controls. Standard errors clustered at the county level in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$